beta

Markets Primer

August 2022

28.1 %
1.1 %
5.5 %
12.1 %
31.2 %
$ 255 636
$ 4 256
$ 52 147
$ 64 369

Contents

Checklist for Preparation 6
Types of Securities 8
Equity 8
Common Stock 8
Preferred Stock 8
Convertible Preferred Stock 8
Debt 9
Fixed Income 11
Money Market Instruments 12
Trading Platforms 13
Exchange 13
Commodity Exchanges 15
Over-the-Counter 16
Exchange Concepts 17
Stock Indices 17
Price Weighting 17
Free-float Market-Capitalization Weighting 17
Exchange Traded Funds (ETFs) 17
Margins 18
Initial Margin 19
Maintenance Margin 19
Clearing House 19
Types of Market Participants 20
Investor vs. Speculator 20
Institutional vs. Retail 20
Trading Concepts 21
Long/Short Positions 21
Bid-Ask Spread 21
Hedging and Speculation 23
Arbitrage 24

Rally / Sell-Off 25
Stop-Loss / Take Profit Order 27
Efficient Market Hypothesis (EMH) 27
Types of Analysis 27
Fundamental Analysis 27
Technical Analysis 27
Fixed Income Concepts 28
Bonds 28
Coupon Bearing Bonds 28
Zero-Coupon Bonds 29
Flexible Rate Notes (FRNs) 30
Tips and Indexed Bonds 30
Treasuries 31
Dirty and Clean Price of a Bond 32
Modified and Macaulay Duration 33
Convexity 34
Credit Rating 35
Yield Curve 36
Types of Yield Curve 36
Yield Curve Trading Strategies 37
Factors influencing Central Bank Policy Decisions 41
Inflation 41
GDP growth 42
Unemployment 42
Factors affecting short term rates 43
Factors affecting long term rates 43
Central Bank Policy Tools 45
Policy Rates 45
Open Market Operations 46
Reserve Requirements 46
Forward Guidance 46

Quantitative Easing (QE) 47
Yield Curve Control (YCC) 47
Derivatives 47
Introduction 47
Spot prices Vs Forward prices 48
Contango & Backwardation 50
Types of Derivatives 52
Forward Contracts 52
Futures 54
Forward Rate Agreement 56
Options 56
Swaps 56
Swaps 57
Introduction 57
Benchmark/Reference Rates 58
Valuation of Swaps 59
Credit Default Swaps (CDS) 61
Overnight Index Swap (OIS) 62
Total Return Swaps (TRS) 63
Options - Basics 65
Introduction 65
Type of Options 65
Factors affecting Option Pricing 66
Put – Call Parity 68
Naked vs Covered Option 68
Vanilla vs Exotic Products 69
Options - Advanced 70
Option Pricing – BSM Model 70
Option Greeks 71
Delta Hedging 73
Volatility Smile 74

Option Strategies 74
Spread 74
Combinations 77
Other Complex Options 78
Futures Options 78
Warrants 79
Embedded Bond Options 79
Interest Rate Cap/Floor 79
Swaptions 80
Securitization 81
Introduction 81
Asset Backed Securities (ABS) 82
Mortgage Backed Securities (MBS) 82
Historical Regulations & Crises 86
Regulations 86
Financial Crises 87

Checklist for Preparation

Technical Knowledge

Good understanding of

- Concepts taught in Financial Markets (FM), Macroeconomics (MEP), Financial Reporting and Analysis (FRA) and Managerial Computing (MC)
- Markets Terminology
- Bond terminology and concepts (PV calculations, Clean Price, Dirty Price, Fixed vs Floating, Credit Risk)
- Forwards, Futures, FRAs, Swaps - Definitions, terminology, payoff diagrams and pricing methodology – Refer Beta Primer and Hull
- Vanilla Options – Payoff Structures, Basics about pricing (Black Scholes), Put-Call Parity and Greeks – Refer Beta Primer and Hull

Macro-View on major economies

- Cover all the recent (past 2-3 years) news regarding major macro events in India (mandatory) and at least 3 other major economies
- Recommended economies: US, Europe, China, Japan, UK, or OPEC as a whole
- Form opinions based on major events in these economies and form view on the financial markets based on the applicability of these opinions in the current scenario

Follow financial markets

- Form opinions and views on the market based on the market news, broad market movement and the extent of correlation between them
- To effectively cover all the major global events, use FT, Bloomberg, The Economist, WSJ and Mint
- Refer to Central bank monetary policy meeting reports
- Start tracking at least 2-3 indices (Nifty, Sensex, S&P, NASDAQ, and Hang Seng, etc.) and/or commodities (Brent/ WTI Crude Oil)
- Following an industry of choice in the equity market can provide a huge advantage
- Use Bloomberg terminal effectively of any market news and reports
- Follow Beta Daily regularly

Puzzles

For preparation of commonly asked puzzles, refer to the preparation folder (Can Dos: Day 1.0 and Heard on the Wall Street) shared by placecom. Further, solve the puzzles sent (nearly 30 to 45 days before the summer interviews) by Abacus through mail with subject “Puzzle a Day”.

FAQs by Interviewers

Spend time on questions like

- Trade Ideas: Follow the news and develop trade ideas based on them. Also develop possible hedges.
- One-year view on any three specific markets
- Why markets?
- Why do you want to join this industry when bonuses are shrinking every day? Where do you think is the future of the business?
- Do you invest? (If not,) How can we assume that you will have a risk appetite while trading for our bank?

Exhaustive list of preparation material

- Read JDs of the companies
- **Technical**
    - Options, Futures and other derivatives – John C Hull
    - BETA Markets Primer
    - Investopedia - to clarify any finance jargon you come across in other sources
- **News Websites:** FT, WSJ, Economist, Bloomberg, Mint
- **Newsletters**
    - FirstFT Asia by Financial Times (provides short summaries of major market news on weekdays)
    - Five Things to Start Your Day – Asia Edition by Bloomberg (similar to FirstFT)
    - The Weekly Fix by Bloomberg (latest fixed income news, charts, and analysis from the Bloomberg Markets team)
    - Finshots Markets and Finshots Daily

Networking

- Talk to as many markets interns as possible to understand everyone's interview/process experience, try to connect most to a Tuccha with similar education / work-ex background as yours.
- Placecom repository for interview experiences

Types of Securities

Equity

A company has two sources of funding - equity or debt. Equity represents the amount of capital invested in the company by the founders/owners. This equity may be divided into stocks, each representing a partial ownership in the company. Stocks are issued to raise capital from the stock markets to fund large scale economic activities, like new projects or expansion. In this case, stocks are issued and people can buy stocks (shares) to become shareholders. A shareholder's liability is limited to the percentage of his ownership or his share in the company.

Types of Equity

Common Stock - This is the most general category of stock. They represent ownership and the associated voting rights in the company proportionate to the number of shares that you have. Any claim on the assets of the company is only residual in the sense that, in case of liquidation, a common stock-holder would be the last one to get paid. The higher risk is compensated by higher returns in terms of dividends and potential appreciation in market price.

Preferred Stock - A preferred stock is of a nature which is in between that of debt and equity. There might be some ownership rights and limited voting rights. The benefits of owning preferred stock is that dividends are usually guaranteed unlike a common stock and they have a preference over common stockholders in case of liquidation.

Convertible Preferred Stock - A convertible preferred stock is generally a preferred stock which is convertible in the sense that it can be exchanged after some period of time (at the discretion of the stock-holder) into common stock.

Example

Take the case of company A which wants to raise capital from the public. The company would float an Initial Public Offering (IPO) by issuing 10 million of shares with face value of Rs. 100. Persons interested in buying shares of this company at a later stage may do so from an "exchange" (say NYSE or BSE). If the demand for this share is very high, the investor might have to purchase the share at a price higher than the nominal value, let us say Rs. 200). The product of market value and number of shares then gives the market capitalization of this company which in this case is Rs. 2 billion.

Applications

A stock issue is generally made by a company when it seeks to obtain huge amounts of fresh capital to fund its economic activities. An investor might go for common stock, preferred stock or convertible stocks depending on his investing needs.

Why do companies resort to equity financing? Why don't they fund their entire needs through debt financing?

The biggest disadvantage of debt financing as opposed to equity financing is that it creates the burden of loan repayment at all times irrespective of whether the company needs more funding (for startup costs etc.) or not. With equity financing, your investors understand the risk of the business and understand that payments (dividends) would be cut during periods when the company makes a loss or when it needs to reinvest funds for growth and expansion. With equity financing, companies do not run the risk of having a bad credit rating (which would make future funding unavailable) if loan repayments are not made on time. Also, since, shareholders have the right to vote in a meeting, they can keep a check on the management and offer significant business insights and value-additions like supplier networks etc. (depending on who your investors are) that the management might not always have.

How can preferred stocks help in preventing hostile takeovers?

Companies can have provisions that allow the directors to formulate the terms and conditions of some preferred shares. In this case, the board would embed a term of a "poison pill" within a preferred share. The 'poison pill' is a warrant or an option that allows holders of the share to get new shares (as a bonus or for substantial discount) if the management of the company changes hand. This would discourage an existing shareholder to acquire more shares as he would have to incur the future liability in the form of potentially significant dilution of his stake. Thus, the acquirer would be discouraged from pursuing a takeover and would have to negotiate with the incumbent board for diluting the provisions of such shares.

Debt

Debt financing is a means for a company (or an individual) to raise finances without diluting the ownership of the entity. It involves borrowing funds from the creditors to meet monetary requirements in return for paying interest on those funds followed by return of principal. The level of interest (which the creditor earns) is proportionate to the risk involved in the application of the money borrowed.

Example

A debt may be issued to meet long term capital requirement as well to meet short term liquidity requirement. Examples of the former would include long term loans from financial institutions and bonds with long term maturities. Examples of the later would include commercial paper and treasury bills (money market instruments). The payment of interest may be semi-annual or annual (typically). For instance, consider a company which takes a bank loan of $100,000 with 10% annual rate of interest for 10 years and principal repayment in the 10th year. The company would get $100,000 to invest in the venture which needs cash. The bank would gain $10,000 in each of the subsequent 10 years and the principal amount of $100,000 in the 10th year.

Applications

Debt financing is considered the chief means of meeting long term capital requirements. This allows the company to obtain funds without transferring ownership (unlike equity financing). The interest payments on many debts are also tax deductible which makes them all the more attractive. Similarly, money market instruments allow a company to meet its liquidity requirements at a low transaction cost and a low rate of interest.

What risks does an investor face when issuing debt to a company?

An investor lending money to a company is typically not actively involved in the management of the firm except in cases of financial distress. Consequently, he faces the risk of the company misappropriating the funds for riskier investments (or other activities unrelated to the contract) as a result of lack of vigilance. They also face the risk of a company taking additional debt and thus increasing its debt/equity ratio which affects the company's financial health. If the latter debt is taken from banks or other senior debtors, then the individual could be subordinated to the new lenders and this would affect his/her repayment expectations during liquidation. Moreover, debts usually offer fixed returns and investors face the risk of a rising inflation that would reduce their real yields. However, there exist various forms of debt such as inflation linked debt and floating debt which can avoid such yield reduction.

What does a high debt-equity ratio indicate?

A high debt-equity ratio indicates that the company has been financing its operations by a relatively large quantity of debt. The exact number which decides whether the debt proportion is excessive or not, varies across sectors. A high ratio would mean that a large part of the company's earnings would go towards servicing the interest on the debt taken. This eats into the portion that remains for the equity holders. In extreme cases, when the earnings are not sufficient to meet the interest cost, the company may get bankrupt leaving potentially nothing for the equity investors.

Fixed Income

Fixed Income instruments, as the name suggests, are financial instruments which give fixed periodic income and may have an eventual return of the principal on maturity (the expiry date of contract). Thus, the cash flows till the maturity of the instrument are known in advance and do not change with changing interest rates, though products classified under this heading do not necessarily reflect this definition anymore.

Example

Government and Sovereign bonds, asset backed securities, corporate bonds (also preferred stock can be considered fixed income instrument) and various derivatives like swaps, options etc. For example, consider a municipal bond issued by the Government of India with a notional value of Rs. 100 and which pays an interest of 5% per annum for 5 years, payable annually. Here, the investor would receive Rs. 5 annually for 4 years and Rs. 105 on maturity, i.e. interest of Rs. 5 and principal payback of Rs. 100 (at the end of 5 years).

Applications

Fixed income instruments are meant for investors who wish to have constant and relatively secure cash flows. This would be useful for investors with less risk appetite, like retired people for whom the stability of assured cash flows is appealing. It is important to note that the present value of the fixed cash flows would change with change in interest rates and other macroeconomic factors. So, an investor with a significant risk appetite and who is looking to take a bet on movement of interest rates or other factors may still invest in these instruments.

Will a person with fixed income return be concerned about rising inflation?

Yes, a person with fixed income returns would be concerned with rising inflation as investors are interested in real monetary returns rather than nominal. Consider for example, that I have Rs 100 invested in an instrument which would make me richer by Rs 10 in the next year (a nominal return of 10%). If the inflation is 10%, then the goods which I can buy for Rs 100 today would cost Rs 110 next year and my 10% nominal return would not allow me to buy more goods and would not make me any richer. If the inflation is more than 10%, then I would end up being poorer in real terms. Thus, inflation is important for persons trading in fixed income securities.

How does such a person hedge himself against this possibility?

A person can invest in TIPS (Treasury Inflation Protected Security) to protect against inflation. These are securities issued by the treasury which change their interest payments as inflation changes. The inflation change is measured by the consumer price index. They can be purchased in increments of $100 (with a minimum investment being $100 and are available with 5, 10 and 15-year maturities.

How do changing interest rates affect the value of a fixed income instrument?

The value of a fixed income instrument moves conversely with the interest rates. For example, consider a bond that was bought at par for $100 and which pays a coupon of 6%. Now, if the general level of interest rates falls (due to deflation or other changes in market conditions), the bond is offering a higher return at par as compared to market rates. Thus, its price would rise and it would start trading above par. The opposite would happen if the general level of interest rates rises.

Money Market Instruments

Money Market Instruments, commonly termed as "paper" are short-term fixed income instruments. These, unlike their long-term counterpart viz. bonds and equities, typically have a maturity of less than one year. A money market is defined as a platform where participants with short term borrowing / lending needs meet and negotiate.

Some examples of "paper" include Treasury Bills (issued by Government), Commercial Paper (issued by companies), Certificates of Deposit (issued by financial institutions), Repurchase Agreements (repo and reverse repo facilities by central banks), etc.

Example

Company A has excess cash on its books which is required for a future investment, say after 6 months. Company B requires cash immediately to meet some obligations. This company is expecting realization of some receivables in next 6 months. Company A can't invest in long terms securities owing to its cash requirement in 6 months. For company B it doesn't make economic sense to issue long term debt and bear high interest rates. Money Market Instruments provide them a platform to meet their short-term liquidity requirements, where company B can issue Commercial Paper which will be bought by Company A.

Applications

The key application of money market instruments is to provide short term liquidity. For corporates the utility of the money market is in terms of working capital management as explained in the example above. In the case of individuals, short term investing opportunities are met by instrument such as T-Bills and Certificates of Deposit.

Why cannot long term liquidity requirements be met by money market instruments?

Money market instruments are designed to meet short-term cash-flow requirements of the issuer. They have maturities of less than a year (typically). As such they cannot be used to meet long term liquidity requirements because issuing them on a rolling basis period after period would involve substantial transaction costs. Although the short-term interest rates are generally lower than the long-term rates (rising yield curve), re-issuing the short-term loan again and again would result in a high reinvestment risk amounting to betting against the forward rates (as they greater than the current spot rate on a rising yield curve).

Why is the yield on Certificates of Deposits higher than that on Treasury Bills?

The yield of a money market instrument reflects the risk that is associated with the instrument. The treasury bills are the safest instruments as they are backed by the US Treasury. Certificates of deposits are promissory notes issued by banks and the risk of repayment is higher for banks as compared to the Treasury. Consequently, certificated of deposits provide a higher yield to compensate for the increase in risk.

What are the other applications of Repurchase Agreements rate (repo and reverse repo)?

Repurchase Agreements rate (repo and reverse repo) are used by the Central Bank to stabilize interest rates. This happens because it affects the cost of borrowing of banks and other registered dealers from the Central Bank and thus the amount of liquidity that banks and financial institutions hold. Money supply can thus be affected, as is the case in India, leading to a change in the real economy interest rates. Repo/reverse repo also acts as a signaling mechanism through which the Central Bank indicates its monetary target to the overall economy.

Trading Platforms

Exchange

An exchange (also known as bourse) is a place where financial instruments are bought/ sold in an efficient and organized manner. The place may be physical or electronic (with the spread of technology, majority of exchange takes place through the electronic medium). The financial instruments may include stocks, options, futures, commodities, securities, foreign currencies and so on.

The transactions in an exchange are usually done with the benefit of a clearing-house (which offers some protection against defaults) as opposed to over-the-counter (OTC) trading which has a greater risk because of the associated uncertainty regarding defaults.

A company which wants to trade securities through an exchange has to fulfill certain listing requirements (the rigidity of which varies across exchanges) like timely disclosures including audited financial reports.

Examples
NYSE (New York Stock Exchange), NASDAQ (National Association of Securities Dealers Automated Quotations), LSE (London Stock Exchange), BSE (Bombay Stock Exchange), NSE (National Stock Exchange), SSE (Shanghai Stock Exchange), SEHK (Hong Kong Stock Exchange), and Euronext

If the default risk is higher in case of OTC trading, why is it more prevalent than transactions through a clearing house in an exchange?

To have its stock traded on an exchange, a company must be listed on that exchange. This means that the company should be able to fulfill the financial requirements for it to become public. The advantages of OTC trading are that you are able to trade in stocks of companies which were unable to fulfill this requirement. Most importantly, in an OTC trade there is a high degree of customization available to the parties involved as opposed to trading on an exchange which usually allow trades in standardized contracts and assets. Another advantage of OTC trading is that transaction costs are lower and thus the effective acquisition price of a share would be lower. In an exchange, the listed companies pay an exchange fee which can be considerable and passed on to the end user can increase the price of the share. With OTC trading, this increase in price is prevented.

How do exchange traded instruments score over OTC instruments in liquidity and price-discovery?

Exchange traded instruments offer better (and fairer) price discovery because the exchange acts as a regulatory authority to protect against price manipulation. Such regulations are however limited in OTC products. Also, the exchange acts as a common ground for all the market participants to converge. This provides an easy mechanism to discover the reservation price of the buyer and the seller and thus the exchange offers better price-discovery. This is also the reason why the standard products of an exchange are more liquid than the customized OTC products. The liquidity for the latter would depend on how many participants are willing to trade on the product which has been tailored in a

particular way. However, some OTC products become "commoditized" as informal standards emerge from convention which increases their liquidity.

Commodity Exchanges

A commodity market is a place (physical or electronic) where market participants can trade in raw materials and primary products. Just like the stock market, there are selected exchanges which allow the investors to deal in standardized contracts and regulate the trading and investment in commodities. Trading in commodities has increased significantly over the recent years. The nominal value of outstanding commodity derivatives has increased by more than 200% on exchanges and by more than 500% in the OTC market.

Examples

The largest commodity exchange in the world is the CME group (NASDAQ: CME) in the United States. Second in line is the Tokyo Commodity Exchange of Japan. The most prominent commodity exchange in India is the Multi Commodity Exchange (MCX) which is the world's sixth largest commodity exchange.

Applications

Commodity markets provide a common meeting ground for people who use or process these commodities and assist them in price discovery. They also help in increasing the liquidity in transactions involving commodities. They are also important for speculators who wish to deal in leveraged instruments and derivatives with commodities as the underlying.

What is the basis risk that a trader in the commodity market would face?

A basis risk can arise because of an imperfect hedge where the asset which is being hedged against does not match the underlying whose product is used to conduct the hedge. In case of commodities, a single product would have huge variations in grades/sub-grades and quality and very often different grades and qualities of the same product can be used as hedges. Consequently, a trader would stand to face significant basis risk. This is also the reason why these markets standardize the exact sub-grade of the asset that is being traded on the exchange.

What are commodity pools?

Commodity pools are “mutual funds” of commodities. They aggregate money from many investors and invest them in commodities. This allows the investor to have limited risks (to the extent of his/her contribution) and still takes part in trades of large sizes. It also allows investors to diversify their risk (like a mutual fund does).

Over-the-Counter

An Over-the-Counter (off-exchange) trading involves the buying and selling of financial instruments without involving an exchange (like NYSE or BSE). The transaction takes place through a dealer network or directly between the buyer and seller.

Example

A large part of the derivatives market is traded in OTC markets. Suppose an investor finds an attractive trading opportunity and is not able to find an appropriate derivative on the exchange. In this case, he can create a new derivative contract and ask for a bid/ask quote from a trader who usually deals in those kinds of derivatives. If both the parties agree on the terms of the contract, the trade is consummated. As this trade did not occur on an exchange or go through a clearinghouse, this will be called an OTC trade.

Applications

Generally, companies which are small, and cannot fulfill the requirements for getting listed in an exchange trade their securities through the OTC route. Similarly, other securities like debt securities and derivatives, which are not traded on an exchange, can be found on the OTC market.

Consider a derivative is traded OTC, who assumes the risk in case of a default? Does the counterparty have any legal resort to protect his interest?

For a derivative traded OTC, the counterparty would assume the risk in case of default. This is the chief disadvantage of trading in OTC markets as compared to trading through an exchange. An exchange has many regulatory mechanisms which OTC trading lacks with relation to counterparty risks.

However, the counterparty can lodge a complaint to the Security Exchange Commission (SEC) or the Financial Industry Regulatory Authority (FINRA) Investor protection in the US. These bodies seek to protect the risk of investors against risks in the OTC markets. Nevertheless, there are companies which are not required to furnish their financial information to the SEC and investors should do more research before investing in such companies.

Exchange Concepts

Stock Indices

A Stock Market Index regularly measures the price performance of a basket of stock from the entire listed stocks on the exchange. Further, it shows the performance of the overall market or a certain segment of the market depending upon the selection of the basket. Often, indices serve as benchmarks against which to evaluate the performance of a portfolio's returns.

The Sensex and Nifty-50 are two popular benchmark indices that largely reflect the performance of Bombay Stock Exchange (BSE) and National Stock Exchange (NSE). To understand how each sector of the stock market is doing, there are sectoral indices such as Nifty Bank. Nifty Auto etc.

The two most common kinds of indices are – Price-weighted and Free Float Market Capitalization-weighted Index.

Price Weighting based indices-based indices weight constituent stocks by its share price. However, a stock split for any constituent stock of the index would cause the weight in the index of the stock that split to decrease, even in the absence of any meaningful change in the fundamentals of that stock. This feature makes price-weighted indices unattractive as benchmarks for passive investment strategies and portfolio managers. Nonetheless, many price-weighted indices, such as the Dow Jones Industrial Average and the Nikkei 225, are followed widely as visible indicators of day-to-day market movements.

Free-float Market-Capitalization Weighting calculates the market capitalization of the company after taking into consideration only those shares of a company that are actively traded in the open market and are not held by the promoters or is locked-in shares in nature. Free float are those shares that are issued by the company which are readily available and are actively traded in the market. Some of the widely used indices include S&P 500, NASDAQ Composite, FTSE 100, NIFTY 50, Sensex, and Hang Seng Index.

Exchange Traded Funds (ETFs)

An exchange traded fund (ETF) is a basket of securities that is traded on an exchange, just like a stock. ETFs offer liquidity anytime through the exchanges. The basket can contain assets such as stocks, bonds or commodities. Most ETFs track an index, such as a stock index or bond index.

Usually, ETFs are passive funds where the fund manager doesn't select components rather simply tries to accurately reflect a chosen benchmark index' performance. And the efficacy of ETFs is measured through the Tracking Error, which measures how closely an ETF

tracks its chosen index. ETFs can be attractive as investments because of their low costs, tax efficiency, liquidity and stock-like features.

The major differences between ETFs and Mutual Funds are:

- ETFs are traded throughout the day on exchanges unlike Mutual Funds which are bought and sold only from the Fund House at the close NAV of the fund.
- Both can track indexes as well, however ETFs tend to be more cost effective (much lower expense ratio compared to mutual) and more liquid as they trade on exchanges like shares of stock.

Margins

One of the ways which contracts are structured in order to reduce counterparty risk is the maintenance of margins by each party to the transaction. Usually in leveraged transactions, an investor's profits or losses are magnified by some factor. To protect the agency which has lent the money for the transaction, the difference between the P&L on the deal and the money lent cannot go below a certain limit. Regular margin postings ensure such requirements are met. There are many kinds of margins such as collateral margin, variation margin, premium margin etc.

Definition

A margin is the required deposit which is a usually a fixed percentage of the total notional value of the financial security that the investor transacting in.

Examples

Collateral Margin - Consider an investor who has recently entered into the short side of a CDS contract wherein he/she provides insurance against the default of a bond. The investor would need to deposit a collateral margin with the counterparty to ensure that the obligations are honored in case the bond does default. Such a margin is called a 'Collateral margin'.

Variation Margin - If an investor has entered into a futures contract, the daily movement of the futures price for the same maturity can lead to profit and loss for the counterparties. The exchange can enforce that such mark-to-market profit and losses are covered by the immediate payment of cash at the end of the day. Such margin is called 'variation margin'.

Premium Margin - The seller of options has the obligation to deliver / buy the security at maturity if the counterparty enforces its right. In order to minimize the default risk of the option seller, a 'premium margin' (equaling the option value) is to be maintained at the exchange. An additional margin, which is generally a theoretical price of an option (could

be the maximum price the option value can attain using historical data or a financial model) less the premium needs to be deposited as well.

Two Major Types of Margins

Initial Margin is the amount of funds required to initiate a position. It represents the percentage of the purchase notional value that must be covered by the investor's own money.

Maintenance Margin is the minimum amount that must be maintained at any given time in your account. In general, the Maintenance margin is less than the Initial Margin. If the funds in your account drop below the maintenance margin level, a few things can happen:

- You may receive a margin call where you will be required to add more funds immediately to bring the account back up to the initial margin level.
- If you do not or cannot meet the margin call, you may be able to reduce your position in accordance with the amount of funds remaining in your account.
- Your position may be liquidated automatically once it drops below the maintenance margin level.

Haircut

Margin doesn't necessarily be the cash. It can be in form of any other financial asset as well. Haircut is defined as the percentage to be deducted from the market price of the financial asset for treating as the collateral. Highly liquid assets such as T-Bills will have a lower haircut whereas illiquid assets will have high haircuts. For example, on pledging of Rs. 100 worth of T-Bills, only Rs. 98 might be lent against the collateral. In this case this extra haircut of Rs.2 creates a situation of overcollateralization.

Clearing House

Financial transactions can take place in an exchange or through an Over the Counter (OTC) market, which is generally used for non-standardized contracts or products. A clearing firm usually exists in both the markets to facilitate the process by acting as the counterparty for both the buyer and seller. The clearing house requires posting of collateral from both counterparties to mitigate its risk. It nets off multiple trades and it helps the market via providing the information of deals executed using its services. This is particularly valuable for OTC products which are generally considered opaque in nature. They also provide the settlement mechanism for trades via which funds & deliverables are transferred.

Example

Some of the trading in OTC products can lead to the buildup of huge positions across several clearing firms which may not be honored in case of adverse movements in markets

causing a huge loss to them. A clearing house can step in to honor the obligations and in return demand collateral from the counterparties.

DTCC (The Depository Trust & Clearing Corporation) is one such example which facilitates the clearing of CDS contracts via netting off of trades of member firms (e.g. if a firm has gone long 100 and short 50 an asset class but with different counterparties, if the clearing house steps in, the firm's net position is only 50 long).

The Options Clearing Corporation is one of the world's largest clearing organizations for options.

Types of Market Participants

The two-basic classification of Financial Market participants are: Investor vs. Speculator and Institutional vs. Retail.

Investor vs. Speculator

Investors - Investors seek to profit through ownership of an instrument by placing greater significance on the underlying fundamental factors such value, growth and size of a firm. An investor is typically distinct from a trader. An investor typically puts capital to use for long-term gain, while a trader seeks to generate short-term profits.

Speculators - Speculation refers to the activity where one takes a position to profit from fluctuations in the market value of a tradable financial instrument—rather than attempting to profit from the underlying financial attributes embodied in the instrument such as value addition, return on investment, growth or dividends. Many speculators pay little attention to the fundamental value of a security and instead focus purely on price movements.

Institutional vs. Retail

Institutional - An institutional investor is an entity which pools money to purchase securities, real property, or other investment assets. In other words, an institutional investor is an organization that invests on behalf of its members and it manages assets based on the interests and goals of its clients.

Institutional investors include banks, insurance companies, pension funds, hedge funds, private equity funds, REITs, endowment funds, sovereign wealth funds, and mutual funds. Each class of Institutional investors differ in their investment horizon time and risk profile.

Retail - A retail investor is an individual who purchases securities for their own personal account rather than for an organization. Retail investors typically trade in much smaller

amounts and exert less influence over corporate decisions than the institutional shareholders.

Retail investing generally occurs through three channels: individual investors, retail brokers (who act at the direction of these individuals), managed accounts (whereby the account manager makes the financial decisions for the individual).

Trading Concepts

Long/Short Positions

Every trade has a long and a short position. Simply put, a long position in a product means that the position holder will benefit if the price of the product or the underlying product (in case of derivatives) increases and vice versa. The opposite is the case for the short position holder

Example

Buying any product is equivalent to taking a long position. Some examples of long positions include buying stocks, bonds, gold, etc. Similarly, selling any product is equivalent to taking a short position. Apart from the above mentioned, other examples include issuing debt, entering into a contract (forward contract) of selling gold at some fixed price, etc.

Can selling a particular product be termed as taking a long position? Is there any such example?

Yes, taking a long position can involve selling a product. For example, consider an investor who takes a long position on a put option which has A as its underlying. This would involve potentially selling the underlying A at an appropriate price and time. In this case, the investor is long the derivative product but short the underlying asset as a result of the relationship between the asset and the derivative.

Bid-Ask Spread

Bid/Ask spread is the amount by which the ask price exceeds the bid price. This is essentially the difference in price between the highest price that a buyer is willing to pay for an asset and the lowest price for which a seller is willing to sell it.

The above definition of bid/ask spread is for a market place. Bid/Ask spread can also be defined for a single financial institution, where bid refers to price at which the financial institution (called as market maker) is ready to buy (or long) an asset and ask is the price at which it is ready to sell (or short) the asset. The traders at investment banks play the

role of Market Makers in this scenario when they are quoting bid/ask spreads on behalf of the financial institution.

Example

Every traded product's price quote (Exchange or OTC) can be given as an example of bid/ask spread. Also, a common example can be a book store that buys old books at Rs 50 per book and sells the same book sat Rs 60 per book.

Taking another example, Bank A is ready to buy Dollars at Rs 46 per dollar while it is ready to sell at Rs 47 per dollar.

Generally, bid/ask spread is lower in more efficient markets. Currency markets as in the above example are clearly more efficient than the old books market.

Applications:

Bid/Ask spread can be thought of as the reward given to the market maker for bearing the risk of maintaining inventory of the particular asset and its associated price fluctuations. Selecting a bid/ask spread for various financial assets is one of the most important decision that a financial institution makes. Too narrow a bid/ask spread will result in lower earnings per cycle whereas an excess wide spread can reduce the number of trades due to the higher friction costs as a result of the high bid/ask spreads. In terms of overall market dynamics, bid/ask spread is an indicator of amount of activity in the market for that particular asset. The more liquid the asset, usually the bid/ask spread would be lower.

Which of these will have a higher bid/ask spread – a blue-chip stock or a small cap company?

A higher bid/ask spread implies that the difference between the price of a buyer and of a seller is larger. This would make it more difficult for them to agree on a price and hence for the trade to execute resulting in low liquidity and volume of trade. The converse is true for a low bid-ask spread. A blue-chip stock would be more liquid as compared to the stock of a small-cap company. The higher volumes of trade in the former would mean existence of a large number of buyers (and sellers) and thus a higher probability for the prices of the two to converge and for the trade to occur. Thus, a blue-chip stock would have a lower bid-ask spread.

What would be the impact of rallies/sell-offs on bid/ask spread?

A rally or a sell off involves a substantially high difference between orders on the two sides of the order book (larger number of buyers in case of a rally and large number of sellers in case of a selloff). Typically, when such a situation arises, liquidity in the asset may dry up as there is a "run-on-the-asset". In such situations, the bid/ask spread might rise sharply reflecting the lack of liquidity in the other side of the trade.

Hedging and Speculation

Speculation refers to the activity where one takes a position in the market with an expectation of the direction and/or magnitude in the movement of the underlying instrument's price.

Hedging is the practice of limiting/minimizing one's risks. It can serve as an effective tool to stabilize the returns one earns on their investments and limits the losses occurring in an unexpected turn of events. It usually involves some cost or giving up some return in exchange for more safety.

Retail investors (to some extent), proprietary trading desks, hedge funds are typically speculators. Some examples of speculation are taking naked positions on options, futures, etc. Hedging can be best understood from the perspective of a business which is exposed to some kind fo market price risk such as foreign exchange risks. In such situations, such businesses can purchase several financial products like futures, forwards and options etc as instruments for the purpose of hedging risks.

Example

Airline A is uncertain about the future prices of fuel and wants to hedge its risk. It can get into a contract of buying fuel at some fixed price on a future date. Such a contract is called a forward contract. Airline A is thus indifferent to changes in the fuel prices since it has locked in its purchase price.

Hedge Fund B, who doesn't need fuel, gets into the similar forwards contract where he agrees to buy fuel at a fixed price on a future date, expecting fuel prices to increase. If the fuel prices actually increase the hedge fund makes money by buying fuel on the fixed price and selling on the market price. However, if the fuel prices dip, the hedge fund makes a loss (In a practical situation though, the hedge fund will square off the contract rather than actually waiting till the maturity date to purchase the oil physically and then sell it.)

Applications

Companies or individuals hedge in order to reduce their risk exposure as mentioned in the above example. One can hedge the input / output prices, returns on an investment, etc. Speculation provides liquidity in the market by many a times forming the counter-party of a hedger.

Is there a natural hedging? What role does vertical integration play in natural hedging?

Natural hedging refers to situations when companies hedge against risks by matching revenues and expenses rather than trading in financial instruments. For example, if a company has its headquarters in country A (faces most of its cost in country A) and sells its products in country B (receiving revenues in B's currency), it faces risk when the exchange rate between A and B changes. The company can hedge itself by allocating some of its costs (say the payments to suppliers, salary bonuses) in the currency of B to match the revenues/payments and cover the risk. This would be natural hedging.

Vertical integration can be used to create natural hedges. For example, consider a retailer. If the wholesale price rises, the retailer faces a risk as he buys at wholesale price. A backward integration into the wholesale business would be a natural hedge against this risk.

What kind of risks should Infosys hedge itself against?

Infosys would hedge itself against changes in the rupee-dollar exchange rates (and against foreign exchange risks in general). This is because majority of its customers are from the US (so its revenues are in dollars) while a large part of its operations is based in India (so its costs are in rupee)

In the above example is Airline A completely indifferent to fuel prices?

Yes, in the above example the airline is completely indifferent to fuel prices (as it has entered into the contract for the same fuel it uses, there is no basis risk as well). The airline may however miss periods of upside when fuel prices are low and competitors are buying at that low price.

Arbitrage

Arbitrage is a trade where an investor makes risk free profits by entering into multiple transactions. It takes place due to some mis-pricing existing in the market. There can be multiple ways of buying an asset (directly or indirect). Arbitrage occurs when there is a pricing mismatch between different routes taken. Investor thus can go long using one route and short using the other as described in the example below.

Example

Consider an investor with £10 million and the following rates:

£1 = $1.5

$1 = ¥80

£1 = ¥100

Long Side: He can buy $15 million from these. With these dollars he can buy ¥1200 million.
Short Side: He can sell ¥1200 million to get £12 million.

This set of transactions creates an overall risk-free profit of £2 million purely due to pricing anomalies existing in the market.

Application

Arbitrage is a widely used concept in theoretical finance where various assets are priced assuming the absence of arbitrage. The assumption made is that the arbitrage opportunity, if any, will last only for an infinitesimal amount of time in efficient markets. Arbitrage occurs from in-efficiencies in the market and acts as a corrective measure to restore the efficiency in the system.

Can there be an arbitrage opportunity in the market for an infinite period of time? How?

In a perfectly efficient market, an arbitrage opportunity would exist only for a finite period of time as market forces would cause the price differential to get eliminated. For instance, if a certain good is trading at a lower price in place A and at a higher price in place B; arbitrageurs would buy from A and sell at B. This would cause the demand at A to go up, forcing up the price at A to rise and a glut of supply at B pulling down the price at B. The process would continue till prices at A and B becomes equal.

The assumption of efficient markets is important as the process would not occur if there are regulations or other barriers that prevent this exchange from occurring. For instance, consider the above example with an added restriction that the good moving into B would attract a high tariff. If this tariff is sufficiently large it would prevent the trade from occurring and thus prevent the prices from getting equal at A and at B.

Also, if there are high transaction costs in exploiting the arbitrage, then it can continue for an infinite period of time

Rally / Sell-Off

These terms are two sides of the same coin. Both imply an unusual change in the trading volumes of a security. One can be termed as the Bull's paradise and other Bear's. Rally and sell-off are the extreme events; however, their usage is quite common even in the normal day to day movements of the market.

Rally implies a massive surge in the buying activity in a security relative to the selling activity. The number of buy orders exceeds the number of sell orders, and this excess of demand over supply leads to an increase in the prices of these securities. A rally is generally triggered by some positive information which may change performance expectations of the security. For example, Indian stock market rallied by 20% on the Monday after the formation of a stable government.

A **Sell-Off** on the other hand implies that the number of sell orders far exceeds the number of buy orders, and this excessive supply may lead to prices falling down. A sell-off is similarly triggered by negative information. For example, Satyam stock plummeted by 80% due to the sell-off of Satyam stocks in the market after the information of internal fraud was made public.

Example

In the events of increasing risk in the economic environment, demand for risk free investments such as Government bonds increases. This would lead to increase in prices of the Government bond. As the prices rise, the yield to maturity of the bond decreases, say from 3.5% to 3.4%, as their present value increase due to the lowering down of discount rates. This is called a rally. Opposite of this occurs when there is an increase in the rates which results in a sell-off. In case of bonds, usually prices are not quoted and yield to maturity (interest rate) is quoted, so a decrease in interest rates would mean the bonds are rallying, while if the interest rates increase, it is called a sell-off.

Is there any upper cap on how much a stock / index can rally on a particular day?

Yes, there are limits on how much certain stock or an index can rise or fall before trading is suspended for the session (circuit breakers).

Why or why not should there be such a cap?

There should be a cap on the percentage that a stock or index is allowed to rally to prevent excessive speculations. An uninhibited rally would move the prices far away from the fundamentals and would increase the volatility by a large amount. Thus, there should be caps on the amount of rally. Similarly, during an excessive sell-off, there is a general sense of an impending catastrophe which may result in investors dumping excessive stock in the

market. To prevent this panic, there are floors on sell-off which trigger a circuit breaker when breached.

Why does yield go down when demand for bond increases?

When the demand for a bond increases, this implies that more investors are prepared to buy the bond for its current yield. This pushes up the price of the bond and thus reduces its effective yield. For example, if a bond offers a coupon of 5% and its price increase from Rs 100 to Rs 110, then its effective yield decreases as now Rs 5 would be earned for an investment of Rs 110 (instead of Rs 100). Thus, the effective yield decreases.

Stop-Loss / Take Profit Order

A **Stop-Loss Order** is a tool used by traders and investors to limit losses and reduce risk exposure. By using this order, a trader limits his risk in the trade to a set amount in the event that the market moves against him.

A **Take Profit Order** allows investors to set a profit limit in which the trade closes once the instrument reaches the specified price target.

Efficient Market Hypothesis (EMH)

The efficient-market hypothesis states that the price of a financial security reflects all the available information of it, like news, fundamentals etc. The implication of the hypothesis is that it would be impossible to outperform the market.

Types of Analysis

Fundamental Analysis

Fundamental analysis is a method of evaluating securities by attempting to measure the intrinsic value of a stock. Fundamental analysts' study everything from the overall economy, industry conditions, competitive position, company's quality of management, corporate governance standards, and the financial condition of company. Earnings, expenses, assets, and liabilities are all important characteristics to fundamental analysts.

Technical Analysis

Technical analysis is a method of evaluating securities by analyzing the statistics generated by market activity, such as past prices and volume. Technical analysts do not attempt to measure an instrument's intrinsic value, but instead use charts and other tools to identify patterns that can suggest future activity.

The field of technical analysis is based on three assumptions – the market discounts everything (at any given time, an instrument's price fully reflects everything that has or could affect the instrument), price moves in trends, history tends to repeat itself.

Fixed Income Concepts

Bonds

Coupon Bearing Bonds

One of the most common way firms raise money from the markets is by issuing bonds. They are interest bearing instruments which promise to pay the bearer of the bond a specified principal on maturity. Coupon bearing bonds pay regular coupons till maturity apart from paying the principal in the end. The coupon rate is simply the ratio of the interest paid to the face value of the bond. Hence, a bond paying a coupon rate of 5% on a face value of 1000, pays 50 (or 25 if the payment is made semi-annually) every time period.

Pricing

The Yield to maturity of the bond is described as the discount rate which when applied to the cash flows (as in coupons and principal) originating from the bond, gives its current price. The yield of a bond trading at par equals is coupon rate. If the bond trades above par, its yield would be below the coupon rate and vice versa if the bond trades below par.

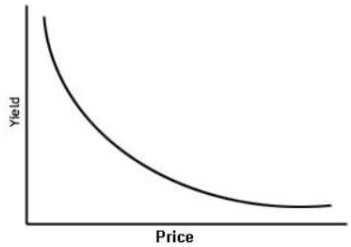
Assume a 5% coupon bearing bond (annual payments) with a face value of 100. The following table gives us the yield when the bond trades at par, premium or a discount.

Trades at		Par	Premium	Discount
End of Year	**Cash Flow**	**PV**	**PV**	**PV**
1	5	4.76	4.81	4.71
2	5	4.54	4.63	4.43
3	5	4.32	4.46	4.18
4	5	4.11	4.29	3.93
5	105	82.27	86.80	77.75
Price of bond:		100.00	105.00	95.00
Yield to Maturity:		5%	4%	6%

The Present value has simply been taken by taking the discounted value of the future value of cash flows (given in the 2nd column) by the appropriate yield to maturity or discount rate (As we described in the section pertaining to Time Value of Money and as given in the last row). The price of the coupon bearing bond maturing after n periods, with r being the yield to maturity and c the coupon payment and FV the principal, is hence:

$$P = \sum_{i=1}^{n} \frac{c}{(1+r)^i} + \frac{FV}{(1+r)^n}$$

Note: The above calculation for pricing a bond assumes a uniform interest rate for all periods. In practice, there are different interest rates pertaining to different periods of maturities. It also assumes that pricing is being done exactly a time period before the coupon payment date.



Price-Yield Curve

Example

Most of the corporate bonds and Government medium- and long-term bonds are coupon bearing bonds which pay annually or semi-annually (In the case of U.S government securities, the payment occurs semi-annually).

Risks

Bonds have several risks associated with them, namely credit risk – the event when the issuer's rating is downgraded, interest rate risk – when interest rates rise the price of the bond reduces, reinvestment risk – in a low interest rate environment the coupons received fetch lower returns in the future, default risk – the chance that the issuer goes bankrupt or cannot service the bond coupon/principal payments in full.

A bond's value is obtained after pricing in all the above-mentioned risks. Two coupon paying bonds may have different risk profiles based on their coupon values and borrower's profile.

Zero-Coupon Bonds

One type of commonly issued bonds are zero-coupon bonds which do not pay any coupon in the period till its maturity, when it pays a promised value of the principal. Naturally these bonds trade at a discount and the difference in the principal and its price is the interest the bond holder earns for the holding period.

A zero-coupon bond can be created by treating each promised cash flow of fixed-income securities as a separate bond itself. Treasury notes are often 'stripped' into different securities which are traded separately. For instance, a treasury note maturing in 5 years paying semiannual coupon payments can be stripped into 11 different securities (10 for

each coupon payment and the principal payment). Such bonds are called STRIPS (Separate Trading of Registered Interest & Principal Securities).

Pricing

Assume a zero-coupon paying bond which pays a principal of 'FV' at maturity. Assuming time 't' till maturity, let the discount rate be 'r' compounded continuously. The price of the zero-coupon paying bond is hence: $P = FVe^{-rt}$

As the equation shows, as the time to maturity decreases, the price of a Zero should approach its principal value.

Applications

Treasury bills are the most commonly found zero coupon bonds which mature within a year. Zeros are also favored by insurance companies as their high interest rate risk act as a hedge against their long-term liabilities. The prices of Zeros are used to construct the yield curve for different maturities.

Risks

Most zero-coupon bonds have short maturities due to the high interest rate risk they carry. This is because, a slight change in the interest rate, can change the price of the bond by a large extent. Since, a Zero-coupon paying bond has no intermediate coupon payments, the investor is also exposed to high default risk and credit risk as well.

Flexible Rate Notes (FRNs)

Unlike regular bonds that pay a fixed rate of interest, floating rate bonds have a variable rate of interest. The coupon rate is reset at a regular interval and is linked to a benchmark rate.

Coupon rate = Benchmark rate + a Fixed Spread

Risks

FRNs carry little interest rate risk. An FRN has a duration close to zero, and its price shows very low sensitivity to changes in market rates. When market rates rise, the expected coupons of the FRN increase in line with the increase in forward rates, which means its price remains constant. Thus, FRNs differ from fixed rate bonds, whose prices decline when market rates rise. The risk that remains is credit risk.

Tips and Indexed Bonds

Bonds whose principal or coupon is indexed against an economic indicator are called 'Indexed Bonds'. Mostly, bonds are indexed to Inflation and are called Inflation-indexed bonds. These securities allow investors to get a real rate of return by hedging themselves against the risk of high inflation. The first known inflation indexed bond was issued by the Massachusetts Bay Company in 1780. Today, the market for inflation indexed bonds is largely dominated by government securities. The U.S issued inflation linked securities are called '*TIPS*' (Treasury Inflation-Protected Securities).

The principal of these bonds is adjusted each year due to inflation based on the Consumer Price Index. A notable feature of these bonds is that the Principal is never adjusted downwards in the instance of deflation.

Inflationary expectations can be implied from the prices at which TIPS trade in the market. For instance, sometime in October 2010, TIPS were auctioned off at a –ve yield (around -0.55%) for the first time ever, which indicated investors' concerns about inflation following the Federal Reserve's quantitative easing policy measures.

Calculation

Assume a $1000 TIPS issued at the beginning of the year with a coupon rate of 5% paid annually. Say the CPI is currently at 100. At the end of the year, if the CPI is 110, the principal of the bond is adjusted by the % increase in CPI, in this case 10%. Hence, the Principal value now is $1100 and the coupon payment at the end of the year would be $55.

Treasuries

Money raised by governments to finance their fiscal schemes is done through Government bonds or Treasuries (in US). These bonds are perceived to be 'risk free' owing to the very low credit risk associated with them (when issued in local currency). The minimum face value of treasuries is $1000. These can be classified into T-Bills (maturing within a year), T-Notes (1 – 10yr maturity) or T-Bonds (> 10yr maturity). The yield curve associated with these securities form the benchmark curve against which all other securities get priced by applying an appropriate spread. The U.S Government also issues Treasury Inflation-Protected Securities, which investors can hold to hedge their risks against inflation.

Applications

The price of sovereign bonds can be an indicator of the country's credit risk. Of late, due to the European sovereign crisis, bonds of Spain, Portugal and Greece among others were trading at a deep discount owing due to the high credit risk associated with them. Government bonds issued in foreign currency (such as the euro or the $) have currency risks associated with them as well.

Dirty and Clean Price of a Bond

As introduced before, the pricing of a Coupon paying bond is done by discounting the cash flows at each period by the yield to maturity. An important assumption taken then was that the pricing is done exactly a period before the coupon payment occurs.

Consider a situation now, where the pricing is done in the time interval between 2 coupon payments. The interest that has accumulated since the previous coupon payment is called 'Accrued Interest'. The bond holder is entitled to receive that interest if he sells it.

The market value of the bond is hence (approximately), the price as calculated from the next coupon payment date plus the accrued interest. On calculating the price as on the next coupon date, we get the bond's '***Clean Price***'. The actual selling price is the bond's '***Dirty Price***' which factors in the accrued interest in between the two coupon paying dates.

$$Dirty\ Price = Clean\ Price + Accrued\ Interest$$

Example

Assume there are N days between 2 coupon payments (C) and n days left for the next coupon date. The accrued interest (using the actual/actual day count convention) is thus:

$$AI = C * \frac{n}{N}$$

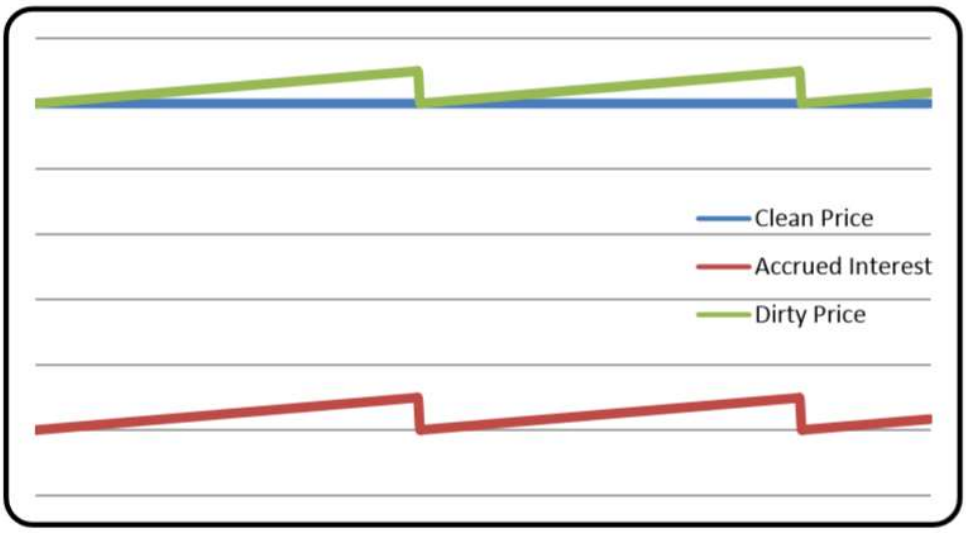
Assume a bond which pays a coupon rate of 10% semi-annually on March 31st and September 30th each year. Let the clean price of the bond as on July 5th be 50. The accrued interest as on July 5th would be

$$AI = 5 * \frac{87}{183} = 2.377$$

The Dirty price of the bond is thus, 52.377.

The accrued interest is at its maximum on the day previous to a coupon payment date. It turns 0 the day the coupon is paid and follows an increasing linear function once again.

The following graph shows the fluctuation of the dirty price as the days proceed (assuming the clean price remains constant at 50):



Applications

Dealers usually quote the clean price of the bond when they trade, though the actual cash exchange happens on the basis of the dirty price. This is because the dirty price keeps changing on a daily basis as the interest accrues. The clean price however reflects the actual pricing of the bond based on the risk factors we have described in the previous sections, and hence is an accurate reflection of the bond's 'value'. For a zero-coupon bond, the clean price and the dirty price remain the same.

Modified and Macaulay Duration

Modified Duration is the sensitivity of a bond or any fixed income instrument to the underlying interest rate. Hence it is the % change in price for a unit change in the underlying discount rate.

Macaulay duration on the other hand is the weighted average maturity of the instrument's cash flows. It's expressed as a unit of time.

Calculation

The price of a bond paying a coupon '*c*' and principal '*P*' at maturity, with a yield to maturity of 'r' would be:

$$V = \sum_{i=1}^{n} \frac{c}{(1+r)^i} + \frac{P}{(1+r)^n}$$

Macaulay duration would be the weighted time average of these cash flows (divided by the price of the bond). Hence, the Macaulay duration would be:

$$Macaulay\ Duration = \frac{\sum_{i=1}^{n} \frac{c*i}{(1+r)^{i}} + \frac{P*n}{(1+r)^{n}}}{V}$$

As is clear from the above formulae, the Macaulay duration would lie between 0 and the instrument's maturity. For Zero coupon bonds, the Macaulay duration would equal the time to maturity for the bond.

Modified duration for the above instrument would be:

$$Modified\ Duration = -\frac{1}{V}\frac{\partial V}{\partial r} = \frac{\text{Macaulay Duration}}{(1+r)}$$

For small changes in the yield, the change in the price would be modified duration times the price of the bond times the percentage change in yield. Hence, if the yield to maturity for a 10-year bond valued at 102 and a duration of 5.34, increases by 0.5%, the price of the bond would fall by 2.73 (5.34 *0.5% * 102).

The duration for higher coupon paying bonds is lesser than a bond paying smaller coupons (assuming both have the same maturity and credit risk). Similarly, a Zero-Coupon bond will have a higher duration than a coupon paying bond with the same maturity.

DV01, another metric used for understanding the sensitivity of fixed income instruments, is the dollar change in the value of a bond for a one basis point change in yield.

Application

Modified duration is an important parameter for investors as it gives an indication about the exposure they are looking at. Usually investors prefer to remain duration neutral in order to ensure volatility in interest rates do not affect them. A variety of instruments such as FRAs or Eurodollar futures are used for this purpose of creating a duration neutral portfolio.

Convexity

Convexity is defined as the 2nd order differential of the price of a fixed income instrument with respect to the interest rate. It is the measure of how the sensitivity of the instrument i.e. duration changes as interest rates rise or fall. One can ascertain the nature of convexity

by plotting the price of the instrument vs. the yield to maturity. A bond with higher convexity will see sharper changes in the price of the instrument for large interest rate movements.

Application

If a position in a bond is duration hedged at a particular point, very small changes in interest rates will result in no loss or gain to the investor. However, if the convexity of the bond is high, large movements in the interest rates can affect the PnL profile.

Bonds with embedded options in it exhibit different behavior. A bond with a call option displays negative convexity at low interest rates as the price of the bond does not change much as interest rates fall as the embedded option imposes a cap on the price even at low interest rates. Similarly, bonds with put options display positive convexity.

Credit Rating

It is an estimate of the credit worthiness of an individual/ organization or a sovereign by a credit rating agency. The credit rating typically depends on some factors like current ratio, debt level, soundness of balance sheet etc. All these factors typically decide the default risk of the entity. Also, different debt issues by the same company might have different ratings.

There is an approximate mapping between the credit spread and the credit rating of an entity. The spread is defined as the excess interest rate over the sovereign rate at which the borrower country might take credit (called the risk premium). Some of the more internationally accepted credit rating agencies (CRAs) are Moody's, Fitch, Standard & Poor and Dunn & Bradstreet.

Example

A bond with a credit rating of BBB and above by S&P is termed as investment grade bond whereas any bond with rating below it is a junk bond. A credit rating system of a particular institution is highly confidential within the institution as that is what that provides them a competitive edge over the others.

Example: Consider two companies IG and JU. IG is a publicly listed company with stable positive cash flows. JU on the other hand is a small privately held company in a cyclical business. JU has more chances of defaulting on the interest payments as compared to IG. Thus, JU will be given a lower credit rating as compared to IG purely based on the line of business. Typically, bonds raised by IG are termed as Investment Grade bonds while those by JU are called Junk Bonds. Demand of junk bonds will be lower resulting in low prices and high yields of the bonds. Credit spread of Junk bonds, thus will be higher as compared to Investment grade bonds, reflecting the risk premium associated with such bonds.

Application

It is not economically efficient and many a times impossible for an individual investor to conduct a proper due diligence on the bond he is investing his funds in, particularly when it forms a small section of his total investment. This is where Credit Ratings come into picture. As mentioned above credit ratings can be mapped with credit spreads and they are usually taken into account by the lenders while fixing an effective interest rate for every borrowing/bond issue.

What will happen to the price / yield of a bond if its rating is downgraded? Upgraded?

A temporary downgrade or upgrade of an issuer's rating may not cause any significant changes in price/yields. However, a series of such events may trigger a response on the price/yield of the bond. For instance, a series of rating downgrades would cause the investor to infer that the issuer's ability to repay the loan is uncertain and will cause the price to go down significantly. Conversely a series of upgrades would cause the price to go up.

Can you give an example where different debts raised by the same company have different credit ratings?

Different debts issued by the same company can have different credit ratings. This is because the rating would depend on the level of seniority of the debt, the type of asset used to secure the debt and the way in which the bond has been structured. An example of this difference would be the debts raised by Bank of Maharashtra. Its lower tier 2 bonds have a credit rating (CRISIL) of AA+ while its tier 1 perpetual bonds have a credit rating of AA.

Yield Curve

Types of Yield Curve

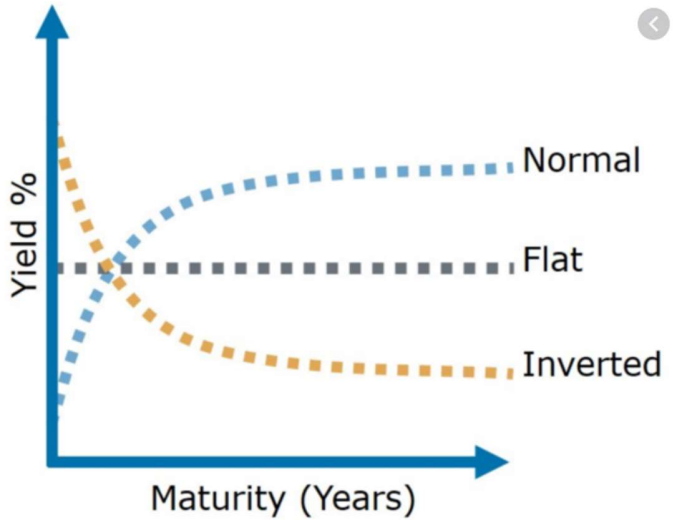
The Yield Curve plots the yields to maturities or interest rates across different maturities for a similar debt instrument (same credit quality and issued by a single issuer). The curve shows the relation between the (level of the) interest rate (or cost of borrowing) and the time to maturity, known as the "term", of the debt. It also reflects the perceived risk of investing in them. Yield Curve is often called as the term structure of interest rates.

The slope of the yield curve gives an idea of future interest rate changes and economic activity. There are three main types of yield curve shapes: normal (upward sloping curve), inverted (downward sloping curve) and flat.

Normal Yield Curve: The market expects the economy to function at normal rate of growth. No significant changes in inflation or available capital. So, investors who risk their money for longer periods expect higher yields.

Inverted Yield Curve: The market expects the economy to slow down and interest rates to drop in the future. Long term investors want to take the opportunity to lock in interest rates before they fall even further.

Flat Yield Curve: The market is at the point of inflection, preceding either a recession or an economic pick-up.



Interpreting the Yield Curve

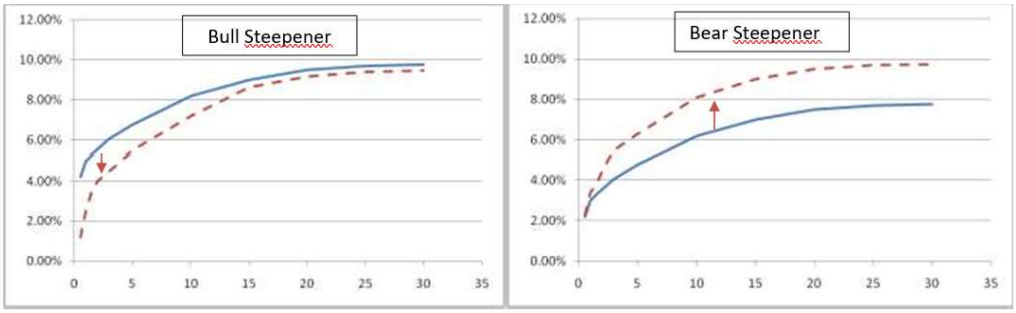
The yield curve is often viewed as a leading indicator, providing an early warning on the likely direction of a country's economy – for example, the yield curve has historically become inverted 12-18 months before a recession in the US.

This is because the slope and shape of the curve reflects investors' expectations about future interest rates, and, by extension, about economic growth.

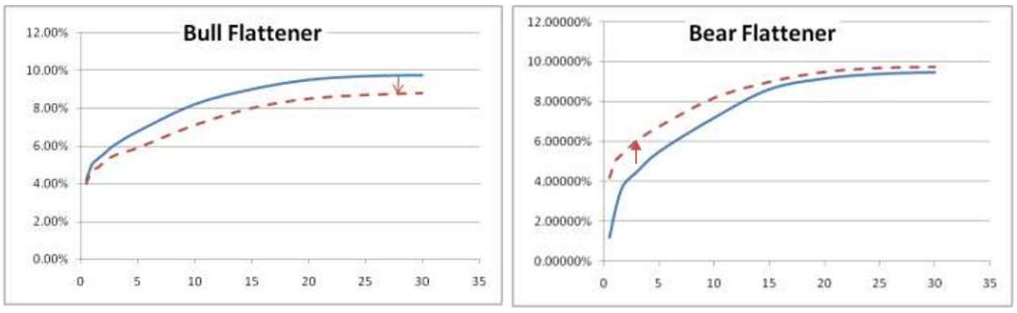
Yield Curve Trading Strategies

A Steepener is a trading strategy which bets that the yield curve will steepen i.e. the gap between the long-term rates and short-term rates will widen. A Bull Steepener strategy bets

on the short- term interest rates falling more sharply than the long-term interest rates. A Bear Steepener, on the other hand, bets that the long-term rates will rise more sharply than short-term rates.



A Flattener is a trading strategy which bets that the yield curve will flatten i.e. the gap between the long-term rates and short-term rates will decrease. In the bull flattener case, the long-term interest rates fall more sharply than the short- term interest rates. In a bear flattener, the short-term rates rise more than the long-term rates.



Steepener		Flattener	
(Long-term rate – Short-term rate) ↑		(Long-term rate – Short-term rate) ↓	
Bull Steepener	**Bear Steepener**	**Bull Flattener**	**Bear Flattener**
Short-term rate ↓↓ Long-term rate ↓	Short-term rate ↑ Long-term rate ↑↑	Short-term rate ↓ Long-term rate ↓↓	Short-term rate ↑↑ Long-term rate ↑

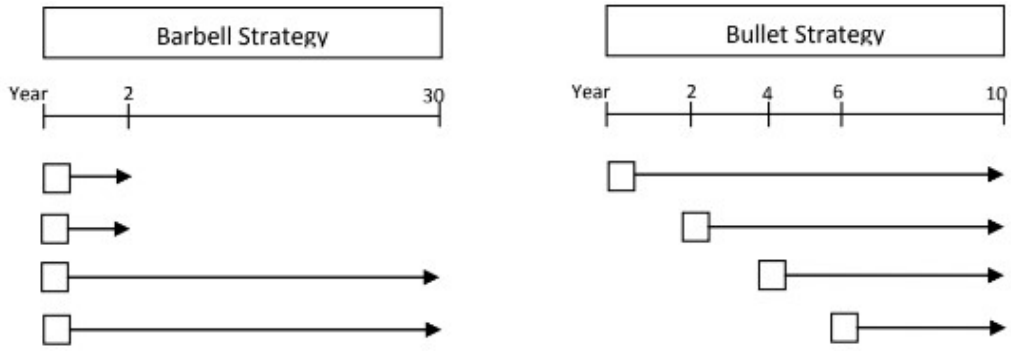
A barbell trading strategy involves holding a portfolio of bonds with maturities at two ends of the yield curve i.e. bonds of very short duration (for e.g. < 2 years) and those of very long duration (e.g.20-30 years). The portfolio consists of bonds concentrated at these two ends of the yield curve and we do not hold any bond with maturity at the middle of the yield curve.

The objective of barbell strategy is to allow the investor to have high-yield from long-term bonds and simultaneously have a part of the portfolio in shorter maturity bonds that are

ready to be deployed when rates rise. When rates fall, the investor enjoys the returns he gets on the longer-term bonds.

When rates rise, the investor reinvests the shorter maturity bonds in longer-term instruments. The objective is to reinvest the shorter maturity bonds into longer-term obligations in such a manner so as to capture the peak interest rates.

The barbell portfolio (with duration D) has much larger convexity than an equivalent portfolio made of bonds with maturity D. Though duration rises linearly with maturity, convexity increases with square of maturity. Hence, A bullet strategy, in contrast to barbell, consists of a portfolio of bonds maturing at the same time but bought at different points in time. Hence, the maturities of the bonds are concentrated at one point on the yield curve. The bullet strategy can be used when we know when we have to pay for our liabilities (college tuition, mortgage balloon payments etc.). However, we stagger the buying of the bonds to minimize the interest rate risk. [convexity of barbell >> convexity of bonds with maturity D]



Example

We can create a portfolio for each strategy as follows: (Please note here that any short maturity bond may be used instead of the 2 yr bond and any long maturity bond may be used instead of the 30yr bond).

Strategy	Portfolio	Alternatively
Bull Steepener	Buy 2 yr. bond Sell 30 yr. bond	Receive fixed 2-year swap Pay fixed 30-year swap
Bear Steepener	Buy 2 yr. bond Sell 30 yr. bond	Receive fixed 2-year swap Pay fixed 30-year swap

Bull Flattener	Sell 2 yr. bond Buy 30 yr. bond	Pay fixed 2-year swap Receive fixed 30-year swap
Bear Flattener	Sell 2 yr. bond Buy 30 yr. bond	Pay fixed 2-year swap Receive fixed 30-year swap
Barbell	Buy 2 yr. bond Buy 30 yr. bond	Receive fixed 2-year swap Receive fixed 30-year swap

For Steepener and Flattener strategy, if the bonds are weighted in the portfolio in such a manner to make portfolio duration zero, the strategy is not affected by small parallel shifts in the yield curve. Hence, a Steepener would make profits if (LT Rates – ST rates) increased and losses if it decreased and vice versa for a Flattener.

Suppose you have to pay $100,000 towards your child's tuition in 10 years' time. You can create a Bullet portfolio as follows:

Year	Portfolio
0	Buy 10-year bonds with face value $20,000
2	Buy 8-year bonds with face value $20,000
4	Buy 6-year bonds with face value $20,000
6	Buy 4-year bonds with face value $20,000
8	Buy 2-year bonds with face value $20,000

Can one create a duration neutral strategy using a barbell and a bullet?

One can bet on the curvature of the yield curve by selling a barbell and buying a bullet. Such a trade is duration neutral and bets on the yield curve to increase at the short and long end of the maturity and decrease at the middle levels.

Under which type of yield curve (increasing/flat/inverted) does it make sense to have a barbell strategy and why?

The barbell strategy makes sense in an upward sloping yield curve as when the 2-year bonds are converted to 30-year bonds, they should capture higher yields. In case of an inverted yield curve, the barbell strategy is expected to underperform a portfolio of only 30-year bonds.

Riding the Yield Curve or Rolling Down the Yield Curve

It is a trading strategy which includes buying and selling of a long-term bond before it matures in order to benefit from the declining yield which occurs over the life of a bond. Investors expect to generate capital gains with the use of this strategy.

As a trading technique, the yield curve performs best in a stable interest-rate setting where interest rates do not rise. In addition, the strategy only produces excess gains if the longer-term interest rate is higher than the shorter-term rate. The strategy provides investors with a higher incremental income without increasing exposure to interest rate risk. Rolling down the yield curve is not a suitable strategy when the yield curve is inverted.

Factors influencing Central Bank Policy Decisions

Different Central Banks have set different targets for themselves which they try to achieve using the Monetary Policy tools like managing short-term rates, cash reserve ratios, open market operations, etc. The major factors that influence the Central Bank decisions are Inflation, GDP growth and Unemployment.

Inflation

As central bank is responsible for price stability; it must regulate the level of inflation by controlling money supplies by means of monetary policy. The central bank performs open market transactions that either inject the market with liquidity or absorb extra funds, directly affecting the level of inflation. To increase the amount of money in circulation and decrease the interest rate (cost) for borrowing, the central bank can buy government bonds, bills, or other government-issued notes. This buying can, however, also lead to higher inflation. When it needs to absorb money to reduce inflation, the central bank will sell government bonds on the open market, which increases the interest rate and discourages borrowing. Open market operations are the key means by which a central bank controls inflation, money supply, and prices. Higher than desired levels of inflation increase the chances of a rate hike by the central bank. Similarly, low levels of inflation invoke a rate cut by the central bank.

What drives expected inflation rates?

Yields, which are essentially Nominal Interest Rates, can change because of real rate changes or because of expected inflation changes.

The economic cycle is intimately linked to investor perceptions of inflation. As the economic growth heats up, firms increase production to meet increasing demand. As production increases, the utilization of existing capacity increases, leading to demands for wage increases as labor becomes scarce. Wage increases induce price increases of

goods, which in turn lead to further wage increases, causing a chain reaction. Due to this chain reaction, consumers and workers expect prices to rise in the future, which increases expectations of inflation in the future.

Another cause of increasing inflation expectation of economic growth is government spending. The government sector has received more attention recently as government expenditures have risen in response to increasing demands for social welfare programs and for government intervention in economic downturns. The first group of spending increases comprises structural government spending mostly unrelated to the state of the economy; the second group of increases is cyclical, rising and falling due to the economy. Excessive government spending can result in increasing amount of debt the government needs to take on which also crowds out private sector debt, leading to interest rate increases.

Real Interest Rates are most affected by market forces of supply and demand. From the supply side, the biggest driver is the perception of risk. For a debt holder, the first major risk is a lack of savings to fund repayment of principal when the time to maturity arrives. For high-grade bonds, such as the US government debt we consider here, the repayment risk is a minuscule one and generally not considered in valuations. Demographic trends, such as aging baby boomers saving for their retirement in the West and similar big shifts in the age structure elsewhere can result in too much savings, increasing the supply of capital. The demand for long term debt is mainly determined by the capital requirement of economic entities which can be aggregated and generalized as the leverage cycle. The term "leverage" refers to the amount of debt versus assets of either an individual or the economy as an aggregate. The leverage cycle has a strong effect on long-term interest rate cycles, with increased borrowing typically leading to higher rates.

GDP growth

The monetary policy decisions taken by the Central bank also depend on the level of economic activity in the economy. If the economic activities have shrunk or are in a declining mode, then the Central bank steps in to spur the economy by adopting an expansionary monetary policy (lower short-term rates, lower reserve ratio, etc.). Likewise, if the economy is overheated, the Central bank steps in to direct the economy towards a stable growth rate by adopting contractionary monetary policy (higher short-term interest rates, higher reserve ratio, etc.) Central banks across the world try to maintain a stable growth in GDP via tools and ensures that economy is not getting into recession or not getting over-heated.

Unemployment

Central banks' decisions also depend on the unemployment levels in the economy. Every economy has a natural sustainable rate of unemployment. If the unemployment levels are persistently below the natural sustainable rate of unemployment, then it can put inflationary pressures on the economy. This would make the central banks hawkish and a short-term rate hike or increase in the reserve ratios are the tools that central bank may look into to pull back unemployment levels to its sustainable levels. Similarly, if the unemployment levels are low that would decrease the prospects of high GDP growth

Central banks are responsible for overseeing the monetary system for a nation (or group of nations), along with a wide range of other responsibilities, from overseeing monetary policy to implementing specific goals such as currency stability, low inflation, and full employment. The role of the central bank has grown in importance in the last century. To ensure the stability of a country's currency, the central bank also acts as the regulator and authority in the banking and monetary systems.

Before we discuss the central banks major policy tools, it is important to understand the factors that affect short- and long-term rates.

Factors affecting short term rates

Worldwide, short-term interest rates are administered by nations' central banks. In the United States, the Federal Reserve's Federal Open Market Committee (FOMC) sets the federal funds rate. Central banks do not control long-term interest rates. Market forces (supply and demand) determine equilibrium pricing for long-term bonds, which set long-term interest rates. If the bond market believes that the FOMC has set the fed funds rate too low, expectations of future inflation increase, which means long-term interest rates increase relative to short-term interest rates – the yield curve steepens.

Factors affecting long term rates

The ability to understand interest rate trends is of interest to policymakers, financial analysts, corporation leaders, and individuals who must make decisions based upon future interest rates. Long-term Interest (LTI) Rates affect key interest-rate sensitive sectors of the economy, such as housing, auto, and investment; interest rates also affect corporate decisions related to pension and asset valuation.

Especially since the beginning of our current financial crisis starting in 2007, it is clear that interest rates impact the economic recovery. Unfortunately, the lack of accurate information related to future trends of LTI impairs the ability of organizations to accurately plan for the future. Policymakers, financial analysts, and organizations need to make specific decisions that are based, in part, on future interest rates trends.

Overnight interest rates, inflation, net capital inflow, budget deficit, and GDP jointly explain, in part, the behavior of long-term interest rates.

Overnight interest rates - Changes (rise or fall) in the overnight interest rate affect bank leaders' decisions regarding borrowing and lending money. For example, a rise in the overnight interest rate discourages bank officials from borrowing, making the rate at which banks lend money high because demand for money will exceed supply for money. On the other hand, falling overnight interest rates motivate bank leaders to be likely to seek and invest in loans. Therefore, the overnight interest rate works like a regulatory instrument that controls the manner in which the U.S. economy works. Changes in overnight interest rate impact long-term interest rates as there is a term premium that the borrower needs to pay on top of the short-term rates so account for the additional risk that a longer-term borrowing possesses.

Expected Inflation - The increase in long-run expectations of inflation leads to higher long-term interest rates as lenders of capital need a higher return to compensate for the increased inflation levels in the future. Similarly, decrease in the long-run expectations of inflation lowers the long-term interest rates.

Foreign capital inflow - Long-term interest rates have been impacted by capital inflow into securities markets. The conundrum surrounding the trend or the pattern of long-term interest rates can be explained by the impact of capital inflow, which has paramount monetary and numerical impact on long-term interest. The foreign purchases of government bonds lead to low levels of long-term interest rates. The negative impact of net inflow on long-term interest rates is an indication that massive capital inflow into a nation results in excess amount of money circulating in the system, which decreases the actual rate of loaning out money and motivates investment.

Budget deficit - A deficit is generated once the government's expenditures exceed its revenues. To finance its operations and increase spending, it borrows money from the private sector and/or from foreign governments. Deficit spending can be leveraged by the federal government as a fiscal policy tool to help stimulate business activity - and hence the economy -- in times of recession. Although increased government spending can create jobs and spur consumer spending, interest rates are bound to rise as the government borrows more. Most of the government borrowings are long-term bonds as they are used to finance the long-term projects and thus the government deficit is associated with an increase in long-term interest rates. Any effort toward lowering the expected level of future national savings places upward pressure on expected short-term interest rates and in turn long-term interest rates as well due to the term premium.

GDP - GDP is the most essential economic parameter as it represents a wide range of economic activity assessment and indicates the pathway of overall aggregate economic activity. In view of this, an increase in nominal GDP results in an increase in spending. Similarly, this means demand for money also must increase to meet spending needs. Should the money supply not increase to meet the demand for money, the result will be higher interest rates, thereby also impacting the longer end of the yield curve.

On a side-note, one topic of recent importance is the Direct Monetization of Deficits.

Direct Monetization of Deficits - A government deficit is said to be directly monetized when the central bank purchases the bonds the government issues to cover its budget deficit in primary market. When government deficits are financed through debt monetization the outcome is an increase in the monetary base and in an expansion of money supply potentially resulting in inflation.

Central Bank Policy Tools

The major policy tools of central banks include Policy Rates, OMOs, Reserve Requirements, Forward Guidance, QE, and YCC.

Policy Rates

Policy Rates the most important policy tool used by the central banks. In the Indian context, the these are the Repo and Reverse Repo Rate.

Repo Rate is the rate at which banks borrow from RBI on a short-term basis against a repurchase agreement. Under this policy, banks are required to provide government securities as collateral and later buy them back after a pre-defined time.

Reverse Repo Rate is the rate RBI pays to banks in order to keep additional funds with the RBI.

Higher the repo rate, more costly are the funds for banks and hence, higher will be the rate that banks pass on to customers. A high rate signals that access to money is expensive for banks; lesser credit will flow into the system and that helps bring down liquidity in the economy.

How do we understanding how much of a cut/hike is priced in the market?

The policy rate set by the Central Bank is a very important rate for a country's economy as it affects monetary and financial conditions. Since most policy rates are one-day rates, the overnight index rate follows the policy rate most closely.

The OIS, or Overnight Indexed Swap, represents a given country's central bank rate over the course of a certain period; in the U.S., that's the Fed funds rate – the key interest rate controlled by the Federal Reserve.

The forward OIS curve is therefore actively monitored to see the market expectations of rate cuts/hikes. Forward dated OIS Swaps between meeting dates are common instruments traded in the market, which indicates the market expected policy rate after each Central Bank meeting.

Open Market Operations

Open market operations (OMOs) are the purchase and sale of government securities in the open market are one of the key tools used by the by a central bank to manage and regulate liquidity on a regular basis. If central banks want to induce liquidity or more funds into the system, it will buy government securities and if it wants to curb the amount of money out there, it will sell these to banks.

Reserve Requirements

In the Indian context, the RBI manages the reserve requirements using CRR and SLR.

Cash Reserve Ratio (CRR) is the percentage of a bank's total deposit that need to be kept as cash with the RBI in the form of reserves or balances. A high percentage means banks have less to lend, which curbs liquidity; a low CRR does the opposite.

Statutory Liquidity Ratio (SLR) is the percentage of net demand and time liabilities (NDTL) that the banks are required to maintain in liquid assets such as cash, gold or RBI approved securities. Banks are allowed to earn interest on these securities. The lesser the amount of SLR, the more banks have to lend outside.

The RBI can reduce or raise CRR/SLR to tighten or ease liquidity as the situation demands.

Forward Guidance

Forward guidance is a tool that central banks use to provide communication either through speeches or meeting minutes to the public about the likely future course of monetary policy and economic outlook. When central banks provide forward guidance, individuals and businesses will use this information in making decisions about spending and

investments. Thus, forward guidance about future policy can influence financial and economic conditions today. In essence, central banks use forward guidance to steer market expectations of future monetary policy moves.

Over the year, the guidance became more explicit, sophisticated, and aggressive from most of the major central banks. For example, the Bank of England tied rate guidance to economic conditions, and the European Central Bank has used guidance to help the public understand the relationships among its diverse set of policy tools.

Quantitative Easing (QE)

Quantitative easing is an unconventional monetary policy and refers to large-scale asset purchases conducted by a central bank with newly-created bank reserves. QE increases the money supply in order to put downward pressure on market interest rates and to expand the economic activity. It also expands the central bank's balance sheet.

It is usually used when inflation is very low or negative, and conventional expansionary monetary policy has become largely ineffective. QE can help bring the economy out of recession and help ensure that inflation does not fall below the central bank's inflation target. One of the risks of QE include inflation overshooting above the target.

Yield Curve Control (YCC)

Yield curve control is also sometimes referred to as yield curve targeting or yield curve caps. YCC aims to control interest rates along some portion of the yield curve. Simply put, YCC imposes interest rate caps on particular maturities. Because bond prices and yields are inversely related, this also implies a price floor for targeted maturities. If bond prices (yields) of targeted maturities remain above (below) the floor, the central bank does nothing. However, if prices fall (rise) below (above) the floor, the central bank buys targeted-maturity bonds—increasing the demand and thus the price of those bonds.

Yield curve control is different in one major respect from QE which is that QE deals in quantities of bonds to be purchased and YCC focuses on prices of bonds.

Derivatives

Introduction

When an investor buys a security or for that matter makes any kind of investment, he bears various kinds of risks. Take an example of the investor, John buying an orange farm in Florida. He is betting on things such as weather conditions in Florida, the prices of oranges etc. He is confident about his farming skills, but he has no way to forecast the future prices of oranges and weather condition in Florida. He wants to do the farming, but without the

risks involved in the business. Weather conditions and variability in oranges prices in future are the main cause of risk in this context. This is where derivatives come into picture.

Definition

Derivatives, as the name suggests are the financial contracts between two counterparties that derive their value from some underlying asset. The underlying can be a stock, interest rate, bond, commodity, currency or anything under the sun that can be measured. (In strict sense interest rate is not an asset, but there are some derivative products such as interest rate swaps which derive their value from interest rate movement.) For example, trading of weather derivatives is quite common in United States. The value of the derivative is a function of the value of the underlying.

Derivatives basically fall into two categories - Forwards (includes forwards, futures, swaps, etc.) and Contingency Claims (includes options).

Applications

Two main applications of derivatives are hedging and speculating. There is no clear demarcation between hedging and speculating. General understanding is that one who owns the underlying is hedging his risk against unfavorable price movements, whereas the one who enters into a position (long / short) without holding a position in the underlying is said to be speculating.

We will continue the aforementioned example of John and try to hedge his risks by using the various kinds of derivatives.

John is bothered about the prices of the Oranges 2 years down the line, when his crop will ripe. He wants to fix the price that he will get by selling his Oranges. He meets a merchant who is looking to buy Oranges two years down the line. He negotiates for the price and gets into a contract to sell him the oranges on a date two years later. This is a simple example of a forward contract.

Spot prices Vs Forward prices

The spot price of any asset class, a commodity such as Gold, Oil etc., or a fixed income instrument such as a bond or an equity stock, is defined as its current market price. This essentially translates to the latest traded price. This price is determined by the market demand and supply of the actual asset.

Forward price is a price at which the buyer and seller agree to exchange the asset at some specified future date. This price is determined by the market expectation of supply and demand during the period before expiry.

However, these two prices are not independent of each other, as we will find out later.

Arbitrage free world

Consider the spot price of an ounce of gold to be $1370 and the risk-free rate (or T-Bill rate) to be 3%. Also, assume that the forward price of gold for a contract with an expiry a year later is $1450.

In such a condition, an investor can borrow $1370 from market at risk-free rate (3%) and buy an ounce of gold with the same. At the same time, he might enter into a forward contract to sell an ounce of gold one year down the line for $1450 (in other words, take a short position in 1-year Gold forward).

After one year, he will get an amount of $1450 by exchanging gold (already purchased) and will have to pay the lender an amount equal to $1370*(1+.03) = $1411.11 assuming annual compounding. Thus, he will make a risk-free profit of $38.9; this is also called pure arbitrage profit.

However there are no risk-free profits in an arbitrage-free world, since more people would be willing to enter into contracts to avail of this arbitrage opportunity and thus the forward price will have a downward pressure and similarly the spot prices will have an upward pressure (as more and more investors would be selling contracts and buying gold).

Now, suppose that the forward price of gold with an expiry of one year is $1400. Here, an investor would enter into a forward agreement to buy gold at $1400, sell gold at $1370 put the proceeds into a risk-free investment with 3% annual return.

At the end of the year, investor would have $1370*(1+0.03) = $1411.11 and would purchase the gold at $1400 (from the contract counterparty). By doing this, the investor makes a risk-free profit of $11.11. Hence, there would be opposite pressures on both the prices, till there are no such opportunities.

The relationship

From our analysis, we can see that any forward price above or below $1411.11 leads to an arbitrage profit. This observation can be generalized through the following relationship between the forward (f) and the spot price (s):

$f = s * e^{rt}$

here 'r' is the risk-free rate and 't' is the time period for the forward contract. This relationship is only valid for those forward contracts in which it does not cost anything to hold the underlying and there are no convenience yields from the underlying (dividend, interest etc.)

A more general equation for the same can be written as

$$f = s * e^{(r-k+u-y)t}$$

Where: r is the risk-free rate, k is the yield from the underlying, u is the storage cost of the underlying and y is the convenience yield of the underlying. Convenience yield is described as the benefit one gets from holding the underlying instead of a forward contract on the same.

In simpler terms, assume that an investor wants to buy a stock an year from now at say 1st January 2012, and this stock yields an annual dividend of d% (paid on 31st December). On 1st January 2011, the price of the stock comprises of two components: present value of dividend on 31st December 2011, and present value of dividends henceforth. Now, when the investor has to price the forward contract on 1st January 2011, he would only take the latter part of the evaluation of stock into consideration. The rationale for this calculation stems from the fact that the owner of the forward contract is not entitled to the income (/loss) of the underlying asset in the duration leading to expiry.

Future prices

Future prices can be understood in the similar way. However, we can't derive any formal relationship between the future and the spot prices due to the daily settlement of profits and losses. As we supply/receive cash during the tenure of the contract depending on the movement of underlying price, the future price will depend on the interest rates in future in addition to the current interest rates.

This can also be understood from the fact that a future contract can be taken as an array of forward contracts. Investor enters into a forward contract every day with the expiry at the end of the day. Thus, the prices of the daily forwards will be impacted by the daily interest rates. The future price quoted is the future price that is arrived upon by the supply and demand of the contracts in the market. This price might vary a little from the forward price on the same underlying as a result of the marking-to-market. This variation will be explained in the Advanced module later on.

Contango & Backwardation

The relation between spot and forward-price for commodities can be described by two states – contango and backwardation. Contango is a situation where the forward curve is upward sloping, i.e. the prices for longer-maturity contracts are more than the prices for shorter-maturities. In backwardation, this forward curve is downward sloping.

To understand this relationship, let us, through an example, look into the trade-offs involved when you decide to buy oil futures (1 barrel) of maturity one year instead of buying 1 barrel oil in spot and storing it for use one year later. If the spot-price per barrel is S_0, then you:

- Earn **interest** on S_0 equal to: rS_0

- Save **storage costs** of storing this barrel: uS_0 (where u is the storage cost for $1 worth of oil)

- This component is important for understanding commodity markets. When you decide to buy future contracts instead of physically holding oil, you forgo the convenience which comes from holding the actual commodity. Notice that if you face a sudden demand for oil in your factory, you cannot burn oil futures, you need real oil!! This comfort derived from possessing the physical commodity is measured by the convenience yield (call it c). The *Risk* magazine (November 2006) defines the convenience yield formally as follows:

"the flow of services and benefits that accrues to an owner of a physical commodity but not to an owner of a contract for future delivery of the commodity. This can come in the form of having a secure supply of raw materials and hence eliminating the costs associated with stock-outs."

No-arbitrage theory tells us that the price of the futures contract of maturity t years is:

$$F_t = S_0(1 + r + u - c)^t$$

As can be seen from this formula, depending on the magnitude of the convenience yield, the futures-price can be either greater than or less than the spot price. The conditions for contango and backwardation can be summarized below:

1. **Contango**: $c < r + u$
2. **Backwardation:** $c > r + u$

A backwardation was observed in the Brent crude oil markets recently. The graph (Source: ftalphaville.ft.com) of Brent-crude's forward-prices, shows that shorter maturity contracts are at a higher price than long-dated ones. This was because an oil trader had hoarded a large amount of Brent thereby causing a crunch in the short-term. This raised the convenience yield as people were concerned about the availability of oil in the near term and the utility of possessing physical oil increased. We note that the interactions between demand and supply affect the convenience yield to a large extent.

Hedge Ratio

Hedging is the art of reducing or eliminating financial risk by entering into a transaction that will protect against loss through a compensatory price movement.

Hedge ratio is a ratio comparing the value of a position protected via a hedge with the size of the entire position itself. In futures, it is a ratio comparing the value of the futures contracts purchased or sold to the value of the cash commodity being hedged. In other words, it is the mechanism for calculating the number of options or other derivatives, or amount of currency, needed to hedge against the risk of loss in a portfolio of shares or other derivatives.

Rolling Hedge

The futures market is highly liquid in the near month futures but the liquidity decreases as the term of the future increases. Suppose you want to hedge an asset with a maturity greater than any currently available future. Here, to hedge your asset, you will have to take a position in a future that matures before your asset. At some point in the future, before the maturity of this future, you will close this position and open a corresponding position in a future that matures at the same time or after your asset matures. This is called rolling the hedge.

Rolling Hedge involves closing out an existing position in the near month futures/ options and opening a new position in the further month futures/options.

Rolling of the hedge may be done many times over the life of the asset being hedged.

Under what kind of conditions (Contango/Backwardation) would the rollover of long position in futures incur losses? When would you make a profit from rolling-over?

When the market is in backwardation, a long position in futures would make profits. If the conditions change to Contango, the position will incur losses. The bet here is essentially on convenience yield being greater than cost of carry. When convenience yield becomes less than cost of carry, a long futures position will make losses.

Types of Derivatives

Forward Contracts

A forward is a financial contract between two parties in which one party agrees to buy a fixed amount of the underlying security from the other on a fixed date at some predetermined price. The buyer is said to hold a long position whereas the seller is holding the short position. The fixed date at which the transfer is to take place is called the maturity date and the predetermined price is called the strike price. The cash transaction takes place only at the maturity of the forward.

A forward is a customizable product and thus traded OTC (Over the Counter). A party willing to exit the contract can do so by either getting into an opposite contract with same or some other party.

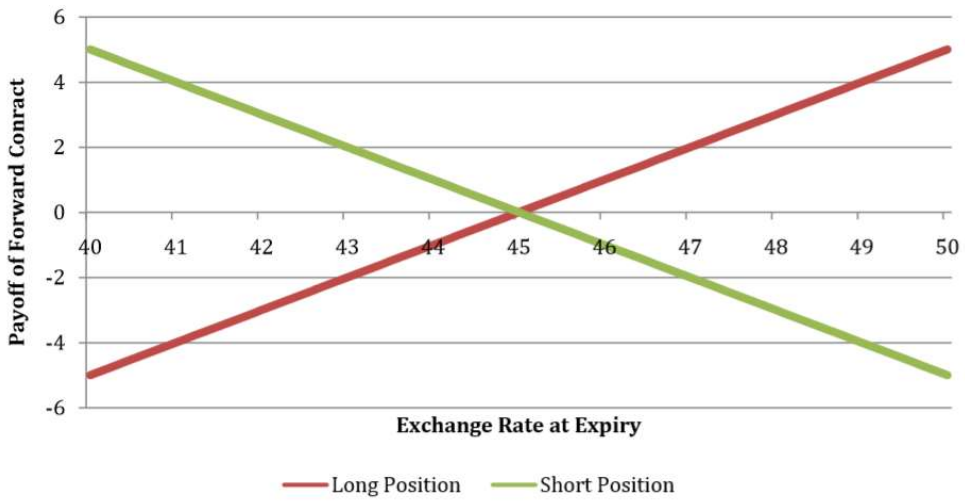
Parameters of a Forward Contract

Following are the parameters that define a forward contract:

1. Underlying: In our example this is Oranges.
2. Strike Price: It is the fixed price at which the delivery is to be made. In our example, it is the price pre-determined by the merchant and John at which the delivery has to be made.
3. Maturity/ Expiry: It is the time when the contract expires. In our example, this is 2 years when the exchange will take place.

Now suppose that the merchant is located in New York whereas John is in Florida. It won't be a viable option for the investor and the merchant to actually exchange Oranges due to high transportation costs, etc. So, they might decide on cash settlement instead of actual exchange of Oranges. At expiry the party holding long position will transfer an amount equal to strike price minus spot price at the expiry date if the actual price is less than the strike price and vice versa. This value is called Payoff at expiry.

Below is the payoff graph for a forward contract for buying Dollars in exchange of Rupees with strike price of Rs 45 per Dollar.



Cash settlement Vs actual transfer of goods

We can show that both of these methods will render the investor and the merchant into situations which are financially equivalent. Suppose the value of underlying is Rs 48 at

expiry whereas the strike price is Rs 45. In case of exchange of underlying, short position holder will transfer the Dollars for Rs 45. On the other hand, in case of cash settlement, the short position holder will pay Rs 3 (Rs 48 – Rs 45). Long position holder will have to buy Dollars for Rs 48, the effective price for him being Rs 45 (he pays Rs 48 for Dollars and gets Rs 3 from the investor).

Using forwards, John has hedged himself against risk associated with price fluctuation in the Oranges market. But, in the process the forward contract has given birth to another kind of risk – risk of default by the merchant. Two years is a long time. During the two years period the merchant might get bankrupt or run away or may simply deny honoring the contract. Such a risk, as discussed below, is called the counterparty risk.

One solution for this is to deal through a mediator who knows both the parties or through some mechanism that can ensure that both the parties will honor the contract. Generally, this is done by an exchange. The exchange plays the role of intermediary for the contracting parties.

To reduce the counterparty risk for the contracting parties, the exchange itself becomes counterparty to the interested parties. Both the parties deal with the exchange on the other side. Thus, both the parties can rest assured that the other party (exchange) will honor the contract (Here we are implicitly assuming that a well-established exchange is more credible than an unknown counterparty). To ensure that the 'party' honors the contract with exchange and does not create additional issues for the exchange, the exchange asks for daily settlement of the prices (Daily settlement is the mechanism used by exchange to reduce its own counterparty risk). Such a contract is called a Futures contract.

Futures

A future is a forward contract with daily settlement of price (A future contract is basically a forward contract with an intermediate exchange for daily settlement of prices). It is a standardized exchange traded product. The standardization is in terms of size of the lot (the number of underlying in one contract) and maturity date. Depending on liquidity (or in other words number of interested counterparties; higher the number easier it is to enter or exit the position), one can enter or exit its position on a future contract. As it is an exchange-traded product with daily settlement of price, there is virtually no risk of default from the counterparty.

In order to hold a position in a future contract, individual has to maintain a margin account (a margin account is explained in detail later) with the broker. Daily settlement of price takes place from the margin account only. If the amount in the margin account becomes lower than a minimum maintenance level, investor gets a margin call (or asked to post

more money in the margin account by the broker). If he/she is not able to do so, the position is squared off by the broker. Futures are available on various underlying assets such as equity, indices, commodities, etc.

Example

Working of margin account in context of a future contract can be understood with the help of the following example.

An investor takes a long position in a future contract trading at Nifty of buying 50 shares of stock 'A' on say 30th September. Spot price of Stock A is let's say Rs 5450 and the future price is Rs 5460. The initial margin is Rs 1000 and the minimum maintenance margin is Rs 500. Below is a set of possibilities.

Day	Action	Price of stock A	Addition to margin amount	Remaining margin
1	Buy	5460	1000	1000
1	Settlement	5470	500(50*10)	1500
2	Settlement	5455	-750(50*-15)	750
3	Settlement	5445	-500(50*-10)	250
3	Margin Call	-	750(paid by inv.)	1000
4	Settlement	5475	1500(50*30)	2500

In the above example, say the price on the expiry date is Rs 5470. In this case, the investor would receive Rs (5470-5460)*50 = Rs 500 from the exchange, who would in turn deduct the amount of Rs 500 from counterparties margin account (margin account makes this transaction virtually risk-free). If any party wants delivery of the underlying, they have to state their intention before the expiry date. Generally, it's the prerogative of the short side to decide the place of delivery (place of delivery is more important in case of commodity futures such as guar, pulses etc.). However, most of the future contracts are settled in cash. In India, National Stock Exchange allows only cash settlement. Our investor, John enters into the futures contract over Oranges. He takes the short position and is not bothered about the merchant. He simply sells a future contract whereas the merchant buys one. The value of the contract at initiation is zero (by our assumption of arbitrage-free world, as we will see later).

Differences between Forwards and Futures

Futures are exchange traded, and thus are virtually risk-free and standardized. Forwards on the other hand are customizable products traded over the counter and thus prone to counter party default risk.

Forward Rate Agreement

Forward Rate Agreements are forward contracts on interest rates, wherein the parties exchange the contracted interest rate vs. the actual interest rate on a pre-determined notional amount. These agreements are bespoke in nature and trade over the counter. Unlike Eurodollar future contracts, they have counterparty and liquidity risks associated with it.

The date on which the reference period for the interest rate begins is called the effective date and the maturity of the swap is termed as the termination date. The underlying interest rate is usually based on a reference rate such as Libor over the period of the swap. Payment on the forward rate agreement occurs on a netting basis. Hence if the actual rate is higher than the contracted rate, the buyer of the contract received an amount equal to the discounted value of the interest differential occurring. The discounting is done since the actual payment though due on the termination date is paid on the effective date.

Calculation

A Trader goes in for an FRA wherein he receives a floating interest rate of 3m Libor and pays a fixed interest rate of 2.5% in return on a notional of $1,000,000. The contract is effective a month from now and terminates after 4 months.

After a month, the 3m Libor rate is declared and its 2.25%. Hence at maturity the fixed rate payer would pay $6,250 and receive $5,625. The difference in the amounts is paid on the effective date. Hence a sum of $621.5 (the discounted value of $625) is paid by the buyer.

Applications

As FRAs are bespoke contracts, they are often used by companies to hedge their interest rate risk. Say a company has issued a floating rate note and needs to hedge itself against the risk that short-term interest rates would rise. In this case it would go short a FRA where it would receive the actual rates and pay the contracted rates, hence eliminating any interest rate uncertainty.

Options

An option, as the name suggests is a financial instrument that provides the holder an ‘option’. In contrast to forward, an option contract provides the holder a right but not the obligation to enter into a trade over the underlying asset (discussed later in detail).

Swaps

A swap is a financial contract to exchange benefits from a series of underlying financial contracts during a fixed period over the duration of the swap. Swaps can be of various types such as currency swap, interest rate swaps, commodity swaps, equity swaps etc. (discussed later in detail).

Swaps

Introduction

A swap is a financial contract to exchange benefits from a series of underlying financial contracts during a fixed period over the duration of the swap. Swaps can be of various types such as currency swap, interest rate swaps, commodity swaps, equity swaps etc.

While entering into a swap the counterparties may (e.g. in case of currency) or may not (e.g. in case of interest rate) exchange the underlying asset (the principal amount). After that for each fixed period, they exchange the returns on these assets. At maturity they finally exchange back the initial assets (only if the underlying was exchanged at the inception of Swap).

Example

Working of a swap contract can be understood with the help of the following example. Consider two companies A and B, which want to take some loan for their financing requirements. Below are the financing options available for Company A and B:

Company	Fixed Rate	Pays by swap
A	6%	LIBOR +2%
B	10%	LIBOR +3%

Now suppose that company A is interested in floating rate and company B is interested in fixed interest rate. Company A takes a loan at fixed rate of 6% whereas B takes a loan at a floating rate of LIBOR+3%.

Now they enter into a swap contract where A pays B a floating rate of LIBOR+2% and 'B' pays 'A' a fixed rate of 7%. The table below summarizes the payments made by them.

Company	Pays to Bank	Pays by swap	Receives by swap	Net Payments
A	6%	LIBOR +2%	7%	LIBOR + 1%
B	LIBOR + 3%	7%	LIBOR + 2%	8%

Thus, as we can see both the parties have benefited by the swap by reducing their effective interest costs.

Types of Swaps

The two most common types of swaps are:

1. Interest rate Swaps – they can be either of 'Fixed Vs Floating' or 'Floating Vs Floating'
2. Currency Swaps – they can be one of these – 'Fixed Vs Fixed', 'Fixed Vs Floating' and 'Floating Vs Floating'

The above-mentioned example is of a fixed Vs floating swap. Other types of swaps can be an equity swap, commodity swap, etc. Swaps are generally Over the Counter (OTC) products.

Benchmark/Reference Rates

A reference rate is an interest rate benchmark used to set other interest rates. Various types of transactions use different reference rate benchmarks, but the most common is the LIBOR, the prime rate, and benchmark U.S. Treasury securities. A common reference rate might be LIBOR, which would be used as a reference in interest rate swaps or an interest rate agreement. In an interest rate swap, the floating reference rate is exchanged by one party for a fixed interest rate or a set of payments. The reference rate will determine the floating interest rate portion of the contract.

Famous Benchmark Rates

LIBOR (Global), EURIBOR (Eurozone), MIBOR (India), MIFOR (India), Fed Funds Rate, TIBOR (Japan), SIBOR (Singapore) etc.

Example

Let's say a homebuyer needs to borrow $40,000 to help finance the purchase of a new home. The bank offers a variable interest rate loan at prime plus 1%. That means the interest rate for the loan equals the prime rate plus 1%. Therefore, if the prime rate is 4%, then your mortgage carries an interest rate of 5% (4%+1%). In this case, the prime rate is the reference rate.

The bank may "reset" the rate from time to time as the reference rate fluctuates. When the prime rate goes up, your rate also goes up. Adversely, when the prime rate falls, so does your payment rate. By allowing the bank to "reset" the rate, it avoids the chance that the borrower may default on the loan, which causes the bank to lose money. Borrowers also benefit from a "reset" of rate. It helps them avoid overpaying for a loan if prime rates happen to go down after the finalization of the loan.

LIBOR

The London Interbank Offered Rate (LIBOR) is a benchmark interest rate at which major global banks lend to one another in the international interbank market for short-term loans. It serves as a globally accepted key benchmark interest rate that indicates borrowing costs between banks. The rate is calculated and published each day by the Intercontinental Exchange (ICE).

LIBOR is calculated for five currencies including the U.S. dollar, the Euro, the British pound, the Japanese yen, and the Swiss franc, and serves seven different maturities—overnight/spot next, one week, and one, two, three, six, and 12 months. The combination of five currencies and seven maturities leads to a total of 35 different LIBOR rates calculated and reported each business day.1 The most commonly quoted rate is the three-month U.S. dollar rate, usually referred to as the current LIBOR rate.

LIBOR Transition

On July 27, 2017 the Financial Conduct Authority, the U.K.'s top regulator, tasked with overseeing Libor, announced the benchmark will be phased out by 2021. Alternative risk-free rates are being set up for the different currencies. For the US Dollar, the US Fed's Alternative Reference Rates Committee (ARRC) has recommended using the Secured Overnight Financing Rate (SOFR), which has been published since April 2019. For the British Pound, it is the Sterling Over Night Index Average (SONIA), and for the Euro, the Euro Short-Term Rate (€STR). Regulators are now expecting financial institutions to actively prepare for the cessation of LIBOR and the adoption of alternative risk-free rates.

Valuation of Swaps

Interest Rate Swaps

An interest rate swap (IRS) is a contract in which two counterparties agree to exchange interest payments at some specified intervals on a specified principal amount for a fixed duration of time.

E.g. Counterparty A will receive 3-month LIBOR and pay a rate of 4% (per annum) to counterparty B every three months based on a notional of $1 million for a total maturity of 1 year. Let us assume this contract was entered on at 1st of January 2011. The exchange of cash-flows is shown below. (Note that the following table has been created after the maturity of the swap; we cannot know the future Libor rates beforehand!)

It is important to note that the principal of $1 million is not exchanged between A and B. However, even if A and B exchanged $1 million at maturity (i.e. 1st Jan 2012), it would not make a difference to the net exchange of cash-flows because the two payments of $1 million from A and B would cancel each other. The cash-flow profile will be the same whether A and B exchange the principal or not. Thus, an IRS can be viewed as an exchange

of a fixed-rate for a floating-rate bond. In this example, A has given a floating-rate bond to B in exchange for a bond, which pays 4% fixed rate per annum.

The above discussion gives rise to a method for calculating the value of an IRS. For counterparty A from our example, the value of the swap (V_A) is simply the difference between the values of the fixed-paying bond (V_{Fix}) which he gets and the floating-rate bond (V_{Float}) which he gives to B.

$V_A = V_{Fix} - V_{Float}$

Generally, for new IRS contracts, the values of fixed and floating bonds constituting the IRS are equal. We know that the floating bond part of the cash-flows will price to par, i.e. 100 (Discussed in detail in the Advanced Module). This means that the fixed bond part must also be priced at 100. In our example, the fixed bond which pays 4% per annum would be priced at 100. This means that the par yield (yield to maturity for a bond trading at par) for bonds, which pay a fixed rate for a maturity of one year, would be 4% (compounded quarterly). Swap rates can thus help us create a par yield curve (i.e. help us calculate the par yields for different maturities). To nail this point, if a newly entered IRS (with a value of zero today) pays a fixed rate of 5% for a maturity of 2 years, we can imply that the par yield for a maturity period of two years is 5%.

Now, suppose a person X wants to enter into the above contract as a fixed rate receiver just after the exchange of cash-flows on 1st April (i.e. just before the LIBOR for the next 3 month period is decided.)

The value of the floating bond at this time is $1 million. For valuing the fixed rate bond, we use the 3- month (3%), 6-month (3.5%), and 9-month (3.75%) Libor rates. For simplicity, assume these are continuously compounded.

The value of the fixed bond comes out to more than $1 million. Thus, X will have to pay counterparty A, a sum of $1,741 to be able to enter this contract as a fixed receiver.

The above discussion brings out two interesting facts. First, the swaps are designed such that their value at inception is generally zero. Secondly, the value of swaps contract changes during the lifetime of swap.

Currency Swaps

A currency swap (CS) is a contract in which two counterparties agree to exchange both principal and interest payments in one currency for those in another at specified intervals for a fixed period of time. A difference between a CS and an IRS is that in a CS, principal is

also exchanged, usually at the first and last payment dates. E.g. On 1st January 2011, counterparty A agrees to receive 5% per annum on $1million and pay 7% per annum on INR 50 million (based on the existing spot rate of $1 = Rs.50) every year for a period of two years. The exchange of cash-flows is shown below:

The following points are worth noting:

1. On the day when the A and B enter this contract, they exchange the two notional amounts. However, as this exchange has been done at the prevailing spot rate of INR 50 per $, effectively the net exchange is zero. Thus we can ignore this exchange of cash-flows while trying to value the swap.
2. Looking at the subsequent cash-flow exchanges, we can see that A has effectively received a $ denominated bond paying 5% per annum on $1 million, and B has received an INR denominated bond paying 7% per annum on INR50 million.

Since we've managed to find an equivalent of the cash-flows of this currency swap in term of two bonds, we can apply the approach we used to value an IRS. Let $V_{\$}$ be the value of the $ bond (in $) and V_{INR} be the value of the INR bond (in INR). If S is the current spot rate in terms of INR per $, the value of this swap to A is:

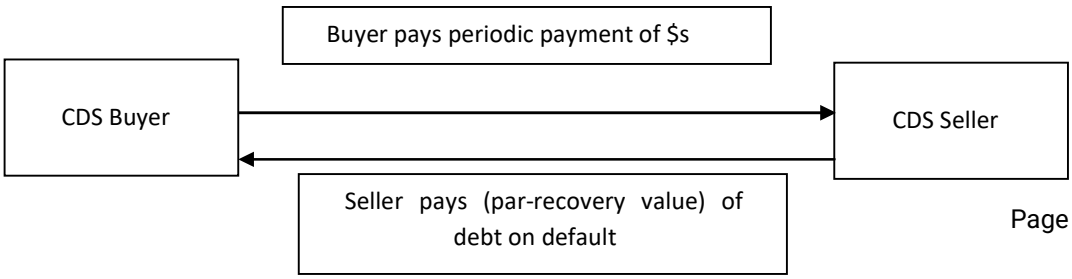
$V_A = S^* V_{\$} - V_{INR}$

($V_{\$}$ and V_{INR} can be calculated using the risk-free rates in $ and INR respectively.)

Credit Default Swaps (CDS)

A CDS is an OTC contract in which one counterparty insures the other counterparty against a default on bonds issued by some specified 'reference entity' for a specified period of time. A CDS is used for hedging a long position in bonds with significant credit risks. Buying protection on a bond simply turns the asset into a credit-risk-free bond. For this reason, the CDS spread for a contract of maturity t years should ideally equal the difference between the t-year par-yield for the reference entity bond and the t-year par yield for a risk-free bond. The CDS spread depends on the probability of default of the reference entity. More the chance of default, greater is the CDS spread.

In a credit default swap, the seller of the swap provides credit protection from the default of some reference entity for periodic payments from the buyer. Essentially, the CDS buyer is buying default insurance from the CDS seller.



CDS buyers are able to transfer the credit risk they face to the CDS seller, thus protecting them in case of default. Also, CDS contracts can be bought by even those who do not have any credit exposure to the reference entity. Buying CDS is equivalent to shorting the reference entity's debt.

CDS sellers gain extra yield by assuming credit risk. Credit risk could also have been isolated by going long the bond and hedging the IR risk. However, that strategy would require funding for buying the bond. Credit risk through selling CDS requires no funding (unless for posting collateral).

Through Default Probability:
We can calculate the CDS premium by estimating the probability of default. The leg where the CDS buyer pays the premium to the CDS seller is called the Premium Leg and the leg where the seller pays the buyer in case of default is called the Default Leg. The CDS premium S is calculated such that
PV [Default Leg] = PV [Premium Leg]

What would an investor wanting credit exposure but having high funding costs prefer – CDS or Asset swap and why?
CDS are mostly preferred by investors who have high funding costs, as it effectively funds the Libor part of the deal for an investor looking for an exposure to the asset.

How would the default contingent IRS exposure affect CDS –cash basis?
The theoretical mark of the Cash CDS is obtained by getting the NPV off the IRS exposure. Hence, if the NPV off the IRS exposure increases, the cash mark increases, reducing the CDS-cash basis.

Overnight Index Swap (OIS)

An Overnight Index Swap (OIS) is an interest rate swap agreement where a fixed rate is swapped against a pre-determined published index of a daily overnight reference rate for example LIBOR for an agreed period. The overnight index swap denotes an interest rate swap involving the overnight rate being exchanged for a fixed interest rate. An overnight index swap uses an overnight rate index such as the federal funds rate as the underlying rate for the floating leg, while the fixed leg would be set at a rate agreed on by both parties.

Note that the swap term is not overnight; it is the reference rate that is an overnight rate. The swap exchanges a fixed term rate for the variable geometric average of the reference daily or overnight rate compounded over the term of the swap. The reference for a daily compounded rate is an overnight rate (or overnight index rate) and the exact averaging formula depends on the type of such rate.

LIBOR-OIS Spread

The LIBOR-OIS spread is the difference between the LIBOR and the overnight index swap rate that is commensurate with credit risk in the interbank lending market. Ordinarily, both the LIBOR and the OIS rates decline when central banks lower their interest rates. However, when the creditworthiness of borrowing banks is in doubt, lending banks charge higher interest rates as compensation for the greater risk. The LIBOR-OIS spread indicates credit risk in the interbank lending market better than the LIBOR itself because the LIBOR is also influenced by the rates set by central banks. Because the overnight index swap rate is based only on the rates set by central banks, subtracting it from the LIBOR shows the risk premium being charged for the credit risk.

Total Return Swaps (TRS)

In a total return swap (TRS), the TRS buyer (TR receiver) receives all the cash-flows generated by a reference entity (e.g. stock or index) from the TRS seller (TR payer). Unlike the Asset swap, here the TRS buyer assumes all risks (credit, IR etc). The main difference between actually buying the asset and buying the TRS is that of ownership i.e. though the asset is never owned by the TRS buyer, s/he assumes all risk associated with the asset and receives all benefits accrued to it.

In TRS, the TRS buyer makes periodic payments to the TRS seller of
(LIBOR + Spread)*(Notional of Reference Asset).

The TRS seller in turn pays out the cash flows (coupons/principal payments/dividends etc) to the TRS buyer.

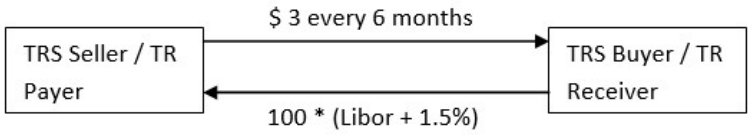
At maturity, in case there is no default and the asset has appreciated in value, TRS Seller pays (Market Price $_{maturity}$- Market Price $_{inception}$). If the asset has depreciated in value, the TRS buyer pays the seller the difference. In case of default, the TRS buyer pays the TRS seller the difference (Market Price $_{inception}$- Recovery Value).

TRS are primarily used when the funding costs required to buy an asset are much greater than the TRS spread quoted on the market. Also, TRS allows investors to short a reference asset synthetically by becoming a TRS seller, which may not be possible in the actual market.

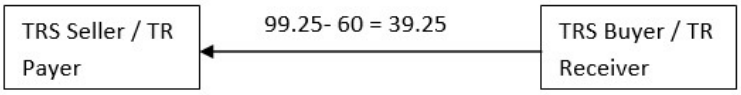
Example

Consider the reference entity is a corporate bond X with coupon rate 6% which pays semi-annually and matures in 5 years. Suppose the current price of the bond in the market is 99.25 and the face value is 100. The TRS swap seller agrees to accept LIBOR + 150bp semi-annual payments in return of the bond's cash flows.

Therefore, during the life of the bond,



If the bond defaults, and recovery value is 60,



If the bond does not default, at maturity, market price is same as par, so the TR payer pays 100-99.25 to the TR receiver.

How can highly rated counterparties use the TRS to take advantage of their low funding costs? What additional risk/s does the TR receiver bear in addition to those of the asset owner?

Highly rated counterparties with low funding costs can act as TRS sellers, where they can borrow at a lower cost to buy the bond and receive higher TRS premiums. However, the TR Payers are assuming the counterparty risk of the TR Receivers in this case. Similarly, the TR Receiver assumes TR Payer's counterparty risk.

Options - Basics

Introduction

An option, as the name suggests is a financial instrument that provides the holder an 'option'. In contrast to forward, an option contract provides the holder a right but not the obligation to enter into a trade over the underlying asset.
There are two parties involved in an option contract. The one who has the right is called option holder and the one who has the obligation is called option writer. Option holder is the one who decides on whether the trade will take place or not. Because of this asymmetric nature of option, holder (buyer) needs to pay some premium to the writer (seller).
Options are fundamentally different from forward and futures contracts. An option gives the holder of the option the right to do something, but the holder does not have to exercise this right. By contrast, in a forward or futures contract, the two parties have committed themselves to some action. It costs a trader nothing (except for the margin/ collateral requirements) to enter into a forward or futures contract, whereas the purchase of an option requires an up-front payment. When charts showing the gain or loss from options trading are produced, the usual practice is to ignore the time value of money, so that the profit is the final payoff minus the initial cost.

Type of Options

There are two most common types of options:

1. Call option: It is a right to buy the underlying asset at the maturity on a predetermined price
2. Put option: It is a right to sell the underlying asset at the maturity on a predetermined price

Such options are called European options. American options provide the right to exercise on any day on or before the maturity.

Options can be traded in exchanges and also OTC. Options can have stocks, bonds, interest rates, currencies, and commodities etc. as underlying securities.

Exercising the option: At the expiry date, if the spot price of the underlying is more (less) than the strike price, then the call (put) option buyer would carry on with the transaction specified, and this is known as 'exercising the option'.

The party which has the right to exercise the option is long the option and party having obligation to pay to option holder is short the option. The latter party is also called the option writer.

Parameters

1. Underlying: Oranges in our case.
2. Strike/ Exercise price: This is price at which the exchange will take place, if the party holding the contract decides.
3. Maturity / Expiry: This is the time period after which (in case of European Options) or in between which (in case of American Options) exchange of underlying take place if the holder of the contract wishes so.
4. Premium: This is the price of the contract (Also called price of option).

Cash Settlement Vs Exchange of Underlying:

An option can result into actual delivery of the underlying or might be settled in cash. Most of the contracts are typically settled in Cash. National Stock Exchange of India makes it mandatory for all the contracts to be settled in cash.

Margin account:

The person who holds the 'option' makes a payment while buying the contract. Subsequent to purchasing the option he does not need to maintain a margin account. This stems from the fact that the holder of 'option' has his risk capped at the price of the 'option'. On the other hand, the writer has non-capped downside as he might have to pay for the contingency claim. Hence the exchange makes it mandatory for him to maintain a margin account.

Pay-Off

The pay-off (the money the option holder would receive on expiry date) at maturity of an option is:

1. Call Option: max (0, S-K)
2. Put Option: max (0, K-S)

Where K is the strike price and S is the spot price on expiry date.

Factors affecting Option Pricing

As the option contract gives the buyer an 'option' but the seller has an obligation, so the seller charges a price for this, known as price of the option. The price of an option depends on following factors:

a) Strike Price: For Call Option, higher the strike, lower the price of the option and vice versa (reverse holds true for Put Option)
b) Spot Price: For Call Option, higher the spot higher the price of the option and vice versa (reverse holds true for Put Option)
c) Volatility of underlying security: Higher the volatility higher the price for both Call and Put Options
d) Time to Maturity: Higher the time to mature higher the price for both Call and Put Options (only if the time to maturity goes very high will this relation reverse because the time-value of future cashflow would reduce; this case can be neglected as there is no liquid markets for these options)
e) Rate of interest: Higher the rate lower the value for both the Options; this comes from the fact that future cash-flows become less valuable (in terms of NPV) than the current outflow of money

Value of the Option

The value of an option is a sum of its 'intrinsic value' and 'time value'. The 'intrinsic value' is the value, which the option would fetch if it were to be exercised immediately. Thus, this is the difference between the strike price and the current spot price of the underlying security. For a call (put) option, the option has positive intrinsic value if the strike price is below (above) the spot price and the option is called being 'in the money'. If the current spot price and spot price are equal, then the option is said to be 'at the money' and if a call (put) option has strike price more (less) than current spot price, then the option is 'out of money'.

Option Styles

- European Option: An option which can be exercised only on expiration date
- American Option: An option which can be exercised on any date on or before expiration date
- Bermudan Option: An option which can be exercised on some specific dates before expiration dates.

For everything remaining the same, since American options give the maximum flexibility and European options the minimum, the following inequality in prices of options holds:

American >= Bermudan >= European

The above inequality can also be interpreted by recognizing the fact that an American option holder has all the right, which European option holder has. In addition to that, he also has few other rights such as early exercise hence the price of an American option cannot be less than an otherwise equivalent European option.

Put – Call Parity

Consider a portfolio consisting of a put option at strike K, maturity = T and a one unit of underlying security.

At t= T, the value of portfolio is:

if S > K: S + 0 = S (the option is not exercised)

if S < K: S + K-S = K (the option is exercised)

The price of this portfolio at t=0 is S_0 + P, where P is price of put option and S0 is spot price of security at t=0.

Now consider another portfolio consisting of a call option at strike K, maturity= T and purchasing a bond which pays K at maturity.

Payoff at t=T for this portfolio:

if S>K: S-K+K = S (option is exercised)

if S<K: 0+K = K (option is not exercised)

Now the payoff of this portfolio is same as that of previous portfolio. So the prices should also be equal at t=0. Thus,

C + PV(K) = P + S

This relation is known as Put- Call Parity.

Note: The put call parity is not applicable to American option as the above derivation assumes that the option cannot be exercised before the expiry. It can be shown that $C_{European} = C_{American}$ and $P_{European} < P_{American}$

Naked vs Covered Option

Naked Option

Naked Call Option is the situation when the call option underwriter does not own the underlying security. In this case, the underwriter is exposed to unlimited risk. The money that the seller would have to pay in case of option being exercised would be the difference of spot price and exercise price, and thus this has no upper cap.

Naked Put Option is similar to naked call option, but the only difference being that in this case the maximum the underwriter may lose is the strike price, as the minimum price of the security cannot go below zero.

Covered Option

Covered Call: When the call underwriter owns the underlying security, the call option is said to be covered.

Covered Put: When the put underwriter has short position (has a negative position) in the underlying security.

The payoff of a covered call is same as that of a naked put and the payoff of a covered put is same as that of naked call. This can be seen by drawing the payoff table or through put-call parity.
By put-call parity,
$C + PV(K)=P + S$
$\Rightarrow S - C = PV(K) - P$
In this, the left-hand side represents a covered call position whereas the right hand side represents uncovered put.
Similarly, $-C = -P-S+PV(K)$
In this, left hand side is an uncovered call position whereas the right-hand side is a covered put position.

Vanilla vs Exotic Products

Vanilla product is an informal but common term used to refer the simplest version of a type of security or financial product. They are much easier to understand for the end customer as compared to their more complex cousins, exotic products. Vanilla products are standardized whereas exotics products are typically customized.

Examples of vanilla and exotic products
Almost all the products traded on the exchange including equity, bonds, futures, simple options come under the category of vanilla products. Exotic products can be created by making some add-ons to the vanilla products. Some examples of exotic products include Collateralized Debt Obligations, Asian Options, Credit Default Swaps, etc. The list of exotic products is endless and is only constrained by the imagination of the individual. One can simply add two vanilla or two exotic products to make a new exotic product.

Example: Simple bond is a vanilla product where the payoff is a fixed interest payment after regular intervals and the principal payment at a pre-defined maturity. This can be made exotic by adding an option that the issuer can call back the bond at some pre-determined price whenever he intends to do so. This product, referred to as a callable bond, is an exotic product.

Applications
Vanilla products are standardized and thus can be traded easily on exchanges. These products have merit when the product has to cater to a large number of participants of different varied financial understanding and sophistication. Exotic products are typically developed in order to cater to the specific needs and traded Over the Counter (OTC).

Option Pricing – BSM Model

The Black-Scholes-Merton model is widely used for pricing European-style derivatives. The model assumes that log returns on the stock price are normally distributed. This property is also known as the lognormal property of stock prices as shown by the following equation

$$\ln (S_T/S_0) \sim \phi[(\mu - \sigma^2/2)T, \sigma^2 T]$$

Assumptions

1) The stock prices are log-normally distributed.
2) Securities can be sold short
3) The markets are perfectly liquid with no transaction costs or taxes and all securities are perfectly divisible
4) There are no dividends during the life of the derivative
5) There are no riskless arbitrage opportunities
6) Security trading is continuous
7) The risk-free rate of interest, mean return of the stock and the volatility of the stock are constant and same for all time periods
8) The derivative can be exercised on the maturity date

Riskless Portfolio

The basic idea behind Black-Scholes-Merton model involves setting up a riskless portfolio that earns the risk-free interest rate in the absence of arbitrage opportunities. Since the derivative price and the stock price are affected by the same underlying, we can set up a riskless portfolio for a short period of time for sufficiently small movements in the stock price. In theory, the portfolio will remain riskless at a given instant and will have to be rebalanced as stock price changes over time.

Once the riskless portfolio has been set up, we claim that the change in the value of the portfolio over an infinitesimally small amount of time will be the interest that it has earned over this amount of time. This is true under the no arbitrage argument.
If the portfolio earned more than the risk-free interest rate then the arbitrageurs would be able to earn a risk-free return by holding the portfolio and borrowing at a risk-free rate of interest. If the portfolio earned less than the risk-free rate of interest, the arbitrageurs would be able sell the portfolio and invest in risk-free securities. In either case, the arbitrageurs would make a risk-free profit.

European Call Option

The boundary condition for a European call option is given by

$$f = \max(S - K, 0) \; at \; t = T$$

where

K – Strike price of the option

The final equation for the price of the call option is given by

$$f = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

where,

$$d_2 = d_1 - \sigma\sqrt{T}$$

European Put Option

The boundary condition for a European call option is given by

$$f = \max(K - S, 0) \; at \; t = T$$

The final equation for the price of the call option is given by

$$f = Ke^{-rT} N - d_2 - S_0 N - d_1$$

Option Greeks

Delta

Delta is a measure of the change in an option's price (that is, the premium of an option) resulting from a change in the underlying security. The value of delta ranges from -1 to 0 for puts and 0 to 1 for calls. Puts generate negative delta because they have a negative relationship with the underlying security—that is, put premiums fall when the underlying security rises, and vice versa.

Conversely, call options have a positive relationship with the price of the underlying asset. If the underlying asset's price rises, so does the call premium, provided there are no changes in other variables such as implied volatility or time remaining until expiration. If the price of the underlying asset falls, the call premium will also decline, provided all other things remain constant.

Delta is commonly used when determining the likelihood of an option being in-the-money at expiration. For example, an out-of-the-money call option with a 0.20 delta has roughly a 20% chance of being in-the-money at expiration, whereas a deep-in-the-money call option with a 0.95 delta has a roughly 95% chance of being in-the-money at expiration.

Gamma

Gamma measures the rate of changes in delta over time. Since delta values are constantly changing with the underlying asset's price, gamma is used to measure the rate of change and provide traders with an idea of what to expect in the future. Gamma values are highest for at-the-money options and lowest for those deep in- or out-of-the-money.

While delta changes based on the underlying asset price, gamma is a constant that represents the rate of change of delta. This makes gamma useful for determining the stability of delta, which can be used to determine the likelihood of an option reaching the strike price at expiration.

A good way to think of gamma is the measure of the stability of an option's probability. If delta represents the probability of being in-the-money at expiration, gamma represents the stability of that probability over time.

Theta

Theta measures the rate of time decay in the value of an option or its premium. Time decay represents the erosion of an option's value or price due to the passage of time. As time passes, the chance of an option being profitable or in-the-money lessens. Time decay tends to accelerate as the expiration date of an option draws closer because there's less time left to earn a profit from the trade.

Theta values are always negative for long options and will always have a zero time value at expiration since time only moves in one direction, and time runs out when an option expires.

Theta is good for sellers and bad for buyers. A good way to visualize it is to imagine an hourglass in which one side is the buyer, and the other is the seller. The buyer must decide whether to exercise the option before time runs out. But in the meantime, the value is flowing from the buyer's side to the seller's side of the hourglass.

Vega

Vega measures the risk of changes in implied volatility or the forward-looking expected volatility of the underlying asset price. While delta measures actual price changes, vega is focused on changes in expectations for future volatility.

Higher volatility makes options more expensive since there's a greater likelihood of hitting the strike price at some point. Vega tells us approximately how much an option price will increase or decrease given an increase or decrease in the level of implied volatility. Option sellers benefit from a fall in implied volatility, but it is just the reverse for option buyers. Long option traders benefit from pricing being bid up, and short option traders benefit from

prices being bid down. This is why long options have a positive vega, and short options have a negative vega.

Rho

Rho measures sensitivity to the interest rate: it is the derivative of the option value with respect to the risk-free interest rate (for the relevant outstanding term). Except under extreme circumstances, the value of an option is less sensitive to changes in the risk-free interest rate than to changes in other parameters. For this reason, rho is the least used of the first-order Greeks.

Delta Hedging

Delta hedging is a type of hedging which involves reducing or eliminating the exposure of a portfolio to small movements in the price of the underlying security. The delta of an option is the first order partial derivative of the option price (O) with respect to price of the underlying asset (S). Delta tells us by how much O will change for a unit increase in S. Using delta, one can calculate a position in the underlying asset which will hedge the option.

Thus, delta hedging an option involves taking an offsetting position in the underlying security/futures. However, this position will have to be adjusted dynamically since delta itself changes as S changes. This is because of the non-linear relation between O and S.

Example

Suppose on Day 1, the stock price is 100 and the premium on a call option to buy 100 shares is 1000. If a trader has sold 5 such options (i.e. options to buy 500 shares), the delta at this point is 0.5 and gamma is 0.1 then,

Position in Option = Number of options owned * Price of one option

*Position in Option = -5 *1000 = -5000*

Delta of the Position of the trader = Number of options owned * Multiplier (Shares/option) * Delta of one option

= -5 * 100* 0.5 = -250

Gamma of the Position of the trader = Number of options owned * Multiplier (Shares/option) * Gamma of one option

= -5* 100 * 0.1 = -50

To make his position Delta Neutral, the Trader will have to buy 250 shares of the underlying stock.

*Position in Stock = 250 * 100 = 25,000*

Suppose on Day 2, the stock price has moved to 101, then Change in Option Position = Delta * (Change in S)

= -250 * 1 = -250

*Position in Option = -5000 - 250 = -5250 (i.e. loss of 250) Position in Stock = 250 * 101 = 25,250 (i.e. profit of 250)*

Hence, the loss made on the Option is offset by profit made in Stocks. New Delta of the Position = Old Delta + Gamma * (Change in P)

= -250 + (-50) * 1 = -300

Hence, the Trader has to buy 50 more shares to make the portfolio Delta Neutral.

Volatility Smile

Volatility smile is the relationship between implied volatility of an option and the strike price for a given maturity where the at-the-money options have lower implied volatility than in-the-money or out-of-the-money options.

Volatility Term Structure

Volatility term structure is the relationship between implied volatility of an option and the time to maturity at a given strike price.

Implied Volatility Surface

Implied volatility surface is a 3-dimensional plot of implied volatility plotted against time to maturity and the strike price of an option. The volatility smile implies that the lognormal distribution underestimates the probability of extreme movements in exchange rates.

Currency Options

Currency options have a volatility smile with volatility increasing towards the extreme ends.

Volatility Skew in Equity Options

The volatility smile for equity options has a skew with increasing implied volatility as the strike price decreases.

A possible explanation for the volatility skew in equity options is the leverage of a company. As the stock price decreases, the value of equity decreases and the amount of leverage automatically increases. As the company's equity becomes riskier, its volatility increases as compared to when the stock price increases.

Option Strategies

Spread

A spread strategy involves positions in two or more options of the same type – for example, either two or more calls or two or more puts.

Bull Spread

A bull spread is created by buying a call option on an underlying with a certain strike price and selling a call option on the same underlying with a higher strike price, with both options having the same expiration date. Using call options, because a call price decreases as the strike price increases, the value of the call sold is lower than the value of the option bought. Thus, a bull spread using call requires an initial investment.

If X_1 is the strike price of the call option with the lower strike and X_2 is the strike of the other call option,

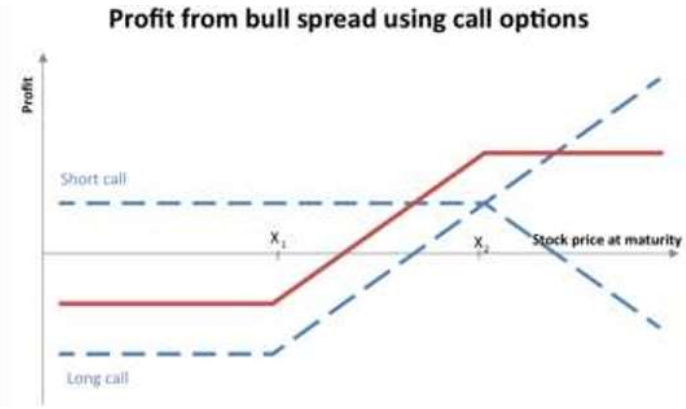


Figure Source: Investopedia.com

Underlying price range	Payoff from long call	Payoff from short call	Total payoff
$S_T <= X_1$	0	0	0
$X_1 < S_T < X_2$	$S_T - X_1$	0	$S_T - X_1$
$S_T >= X_2$	$S_T - X_1$	$X_2 - S_T$	$X_2 - X_1$

Bear Spread

An investor who enters into a bear spread is hoping that the stock price will go down. It can be created by buying a put with one strike price and shorting a put with a lower strike price. A bear spread created using puts involves initial investment since the put price of the option bought is more than the price of the option sold. Like bull spreads, bear spreads limit both the upside profit potential and the downside risk.

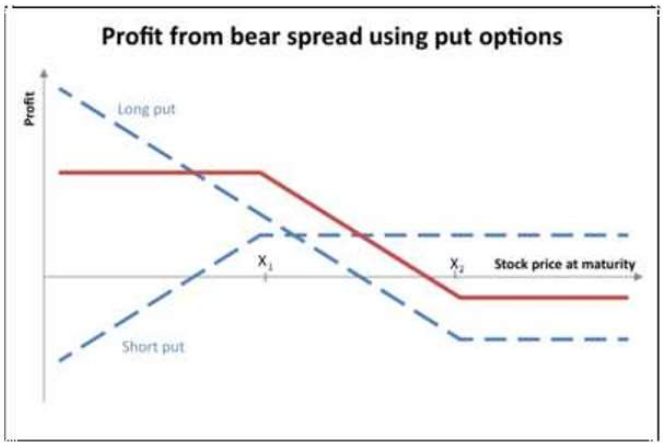


Figure Source: Investopedia.com

Underlying price range	Payoff from long put	Payoff from short put	Total payoff
$S_T <= X_1$	X_2-S_T	S_T-X_1	X_2-X_1
$X_1 < S_T < X_2$	X_2-S_T	0	X_2 - S_T
$S_T >= X_2$	0	0	0

Box Spreads

A box spread is a combination of a bull call spread with strike prices of K1 and K2 and a bear put spread with the same strike prices. The payoff from a box spread is always K2-K1. The value, therefore, of a box spread is always $(X_2-X_1)e^{-rT}$ An arbitrage opportunity exists if the value is different from this.

Buying the box implies buying a call with a strike price of X_1, buying a put with strike price of X_2, selling a call with strike price of X_2, and selling a put with a strike price of X_1.
It is important to remember that a box spread arbitrage works only with European options.

Butterfly spread

A butterfly spread involves positions with three different options (with different strike prices). It can be created by buying a call option with a relatively low strike, K_1, buying a call option with a relatively high strike price, K_3, and selling two call options with the same strike price, K_2, halfway between K_1 and K3. Generally, K2 is closer to the current stock price. The strategy leads to a profit if the underlying price stays close to K2, but gives rise to a small loss if there is a significant movement in either direction. Thus, one enters a butterfly spread if the view on the market is against significant movements on either side.

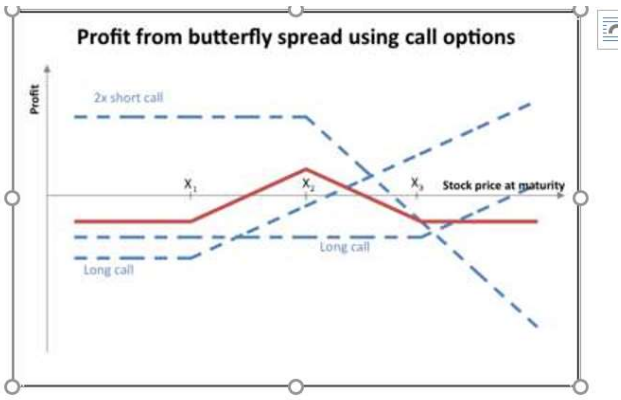


Figure Source: Investopedia.com

Underlying price range	Payoff from bull call spread	Payoff from bear put spread	Total payoff
S_T <= X_1	0	K_2 - X_1	X_2-X_1
X_1 < S_T < X_2	S_T - K_1	K_2 - S_T	X_2 – K_1
S_T >= X_2	K_2 - K_1	0	K_2 - K_1

Combinations

Straddle

A straddle is a popular combination strategy. It involves buying a call and a put with the same strike price and the same maturity. If the underlying is close to the strike price at the time of maturity, then the straddle leads to a loss. However, if there is large movement ion either side, the strategy leads to handsome payoffs.

Thus, a straddle is appropriate when one has the view that there will be significant movement in the market but is not sure of the direction of the movement.

Going long a call and a put to enter into a straddle is called a bottom straddle. A top straddle is the reverse position – going short both a call and a put.

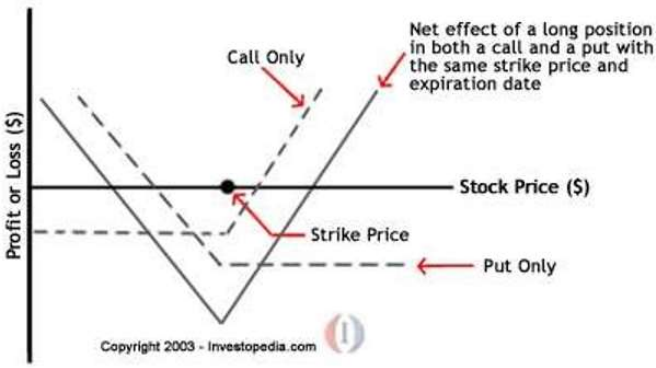


Figure Source: Investopedia.com

Underlying price range	Payoff from call	Payoff from put	Total payoff

S_T <= X	0	K - S_T	K –S_T
S_T > X	S_T - K	0	S_T - K

Strangle

In a strangle, an investor buys a put and a call (or shorts both a put and a call) with the same expiration date, but with different strike prices. In the figure below, the call strike price, K_2 is higher than the put strike price, K_1.

In a strangle, like a straddle, the investor is betting on a huge price movement, but not sure of the direction in which the move will happen. However, as compared to a straddle, the downside risk is lower in a strangle. Also, the price movements need to be bigger in a strangle to be able to make profits. The profit pattern depends on how far the two strike prices are from each other – the farther they are apart, the less the downside risk (and the higher price movement needed to make profit).

Underlying price range	Payoff from call	Payoff from bear put spread	Total payoff
S_T <= K_1	0	K_1 - S_T	K_1 - S_T
K_1 < S_T < K_2	0	0	0
S_T >= K_2	S_T - K_2	0	S_T - K_2

Other Complex Options

Futures Options

Options on futures as the underlying are known as futures options.
A futures option is the right but not the obligation to enter into a futures contract at a specified futures price at a specified time. For example, a call futures option is the right to enter into a long futures contract at a specified futures price at a specified time.
Futures options are usually American.

Expiration

Futures options are identified by the delivery month of the futures contract underlying the option and not by the expiry date of the option itself.
When the underlying futures contract matures at the same time as the option, the European futures options and spot options are the same.

Payoff

The payoff of a European call option with strike price K on the futures contract of the underlying asset is

$$\max(F_T - K, 0)$$

Warrants

A *warrant* is an over the counter (OTC) traded instrument gives the holder the right to buy the underlying security (usually equity) on or before a specific date at a specified price. Warrants are often issued in conjunction with debt to offer a higher yield to the investor. Warrants are like options in that they give the holder the right to buy the underlying security at a pre-specified price and time. However, a warrant is issued and guaranteed by an institution (usually the corporation that issues the debt) while an option is issued and traded publicly on an exchange.

Embedded Bond Options

A bond option is the right and not the obligation to buy or sell a bond by a particular date at a particular price. Bond options are explicitly traded in over-the-counter market or embedded in the bond.

Examples of bonds with embedded options are *callable bonds* and *puttable bonds*.

Callable Bond

A callable bond gives the issuer the right to buy back the bond at specific dates at a pre-determined price. Since the callable bond is an option held by the issuer, the issuer usually pays a higher yield as compared to a non-callable bond.

Puttable bond

A puttable bond gives the bond holder the right to sell the bond at a predetermined price at specific dates. The investor holds the option of selling the bond. Hence, a puttable bond usually earns lower yield as compared to a non-puttable bond.

Interest Rate Cap/Floor

An interest rate cap is designed to provide insurance against the rate of interest on the floating-rate note rising above a certain level. This level is known as the cap rate. Suppose that the principal amount is $10 million, the tenor is 3 months, the life of the cap is 5 years, and the cap rate is 4%. (Because the payments are made quarterly, this cap rate is expressed with quarterly compounding.) The cap provides insurance against the interest

on the floating rate note rising above 4%. For the moment we ignore day count issues and assume that there is exactly 0.25 year between each payment date.
Suppose that on a particular reset date the 3-month LIBOR interest rate is 5%. The floating rate note would require

0.25 X 0.05 X $10,000,000 = $125,000

of interest to be paid 3 months later. With a 3-month LIBOR rate of 4% the interest payment would be

0.25 X 0.04 X $10,000,000 = $100,000

The cap therefore provides a payoff of $25,000. The payoff does not occur on the reset date when the 5% is observed: it occurs 3 months later. This reflects the usual time lag between an interest rate being observed and the corresponding payment being required. At each reset date during the life of the cap, LIBOR is observed. If LIBOR is less than 4%, there is no payoff from the cap three months later. If LIBOR is greater than 4%, the payoff is one quarter of the excess applied to the principal of $10 million.

An interest rate floor is a similar concept, but it allows insurance against rate of interest dipping below a certain level (used to hedge investment income rather than borrowing cost).

Swaptions

As the name suggests, a swaption is an option to enter into a swap at a predetermined time in the future. The option can be of European, American or Bermudan type, where in the last case, the option can be exercised only on predetermined dates within a specified range (the other two cases are normal European and American option scenarios)
Swaption can be of two types

- Payer Swaption: Option to enter into a swap, being the fixed rate payer
- Receiver Swaption: Option to enter into a swap, being the fixed rate receiver

Now consider a payer swaption. Under what condition the option would be exercised? Remember it's an option to pay a fixed rate and receive a floating rate. So, at maturity of the option, if the swap rate turns out to be lower than the strike rate, the Payer Swaption would not be exercised. Therefore, it is akin to a call option on the forward swap rate and the pay-off at each payment date can be formulated as

Max (Swap Rate (t, T) – Strike Rate, 0) X Notional / (No of payments / Yr)

Where Swap Rate (t, T) means Swap rate at time t for the time period t to T

Similar deductions can be made about the receiver swaption as well.

Bond options analogy

We know that a receiver swaption is an option to receive fixed rate and pay floating rate. This can also be thought of as a call option on a fixed rate bond (what is the strike price? Hint think about the present value of the floating leg). While a payer swaption can be thought of as a put option on a fixed rate bond.

From this discussion we can deduce that with all factors same,

Value of (Receiver Swaption - Payer Swaption) = Value of Forward Swap

Difference with a cap/floor

A Cap can be thought of as a set of call options on forward rates (each individual option is known as a caplet). So the pay-off for each caplet would be

Max (Libor Rate (t, T) – Strike Rate, 0) X Notional / (No of payments / Yr)

Where Libor Rate (t, T) means Libor rate at time t for the time period t to T

While a Floor is a similar series of put options (each constituent known as a floorlet).
Now Cap being made up of individual call options, each such constituent Call Option can be exercised independently. This is where a Cap is fundamentally different from a Swaption. In a Swaption, once we exercise the option to Pay (Receive) Fixed, the optionality no longer exists and it becomes a normal interest rate swap with Fixed Rate Vs, Floating rate exchange, whatever be subsequent Swap Rate dynamics. However, in a Cap, we may choose to exercise each Caplet independently from the others.

Securitization

Introduction

Securitization is the financial practice of pooling various types of contractual debt such as residential mortgages, commercial mortgages, auto loans or credit card debt obligations and selling their related cash flows to third party investors as securities, which may be described as bonds, pass-through securities, or collateralized debt obligations (CDOs). Investors are repaid from the principal and interest cash flows collected from the underlying debt and redistributed through the capital structure of the new financing.

Traditionally, banks have funded their loans primarily from deposits. In the 1960s, US banks found that they could not keep pace with the demand for residential mortgages with this type of funding. This led to the development of the mortgage-backed security (MBS) market. Portfolios of mortgages were created and the cash flows (interest and principal

payments) generated by the portfolios were packaged as securities (i.e., securitized) and sold to investors. The US government created the Government National Mortgage Association (GNMA, also known as Ginnie Mae) in 1968. This organization guaranteed (for a fee) interest and principal payments on qualifying mortgages and created the securities that were sold to investors.

Thus, although banks originated the mortgages, they did not keep them on their balance sheets. Securitization allowed them to increase their lending faster than their deposits were growing. GNMA's guarantee protected MBS investors against defaults by borrowers.

Asset Backed Securities (ABS)

As the name suggests, Asset Backed Securities are securities that are backed by specified underlying assets. The underlying assets are typically illiquid assets, like mortgages, credit card loans, student loans that are securitized to create tradable securities that have a claim over cash flows from the underlying assets.

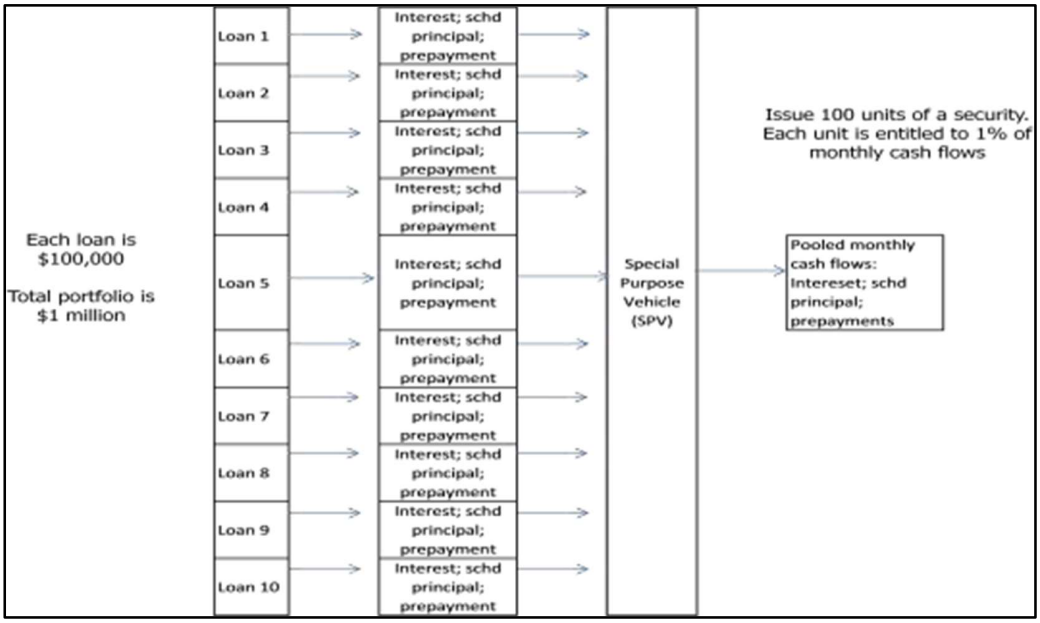
Securitization is a process of increasing liquidity of assets and distributing risk, by aggregating a pool of assets and issuing debt obligations that have a claim over the cash flows from these asset pools.

In this section, we would focus primarily on the ABS with mortgages as underlying assets. The structures of ABS with other underlying assets are similar, although their underlying risk factors and market dynamics would differ.

Mortgage Backed Securities (MBS)

Mortgage-backed securities (MBSs), as the name suggests, are debt obligations that have a claim over cash flows from a pool of mortgages. Securitization of mortgages provides funds to the lenders to issue new mortgages by selling off the existing mortgages to investors.
There are typically two kinds of MBS-

MBS Pass through
Pass-throughs are structured as trusts in which mortgage payments are collected and passed through to investors. They typically have stated maturities of five, 15, or 30 years. The life of a pass-through may be less than the stated maturity depending on the principal payments on the mortgages that make up the pass-through.



Collateralized Mortgage Obligation (CMO)

CMOs are created when the cash flows from an MBS pool are split into multiple tranches. These tranches have priority based principal payments (principal waterfall) and thereby they have differential prepayment and default risks. This enables the investors to assume the prepayment risk in mortgages that suits their risk-return profile.

CMO is the mortgage counterpart of Credit Debt Obligation. In CDO, the primary risk that is distributed is the default risk. In CMO, tranching creates differential prepayment risk- return profile across different tranches.

Principal Waterfall

The most typical CMO structure is sequential principal waterfall. The cash flow structure is as follows:

- All the tranches receive monthly interest on the outstanding balance of the respective tranche
- The monthly principal payment, scheduled and prepayment, are first used to pay the tranche A until it is completely paid off
- When tranche A is completely paid off, the tranche B receives principal payments until it is paid off. As tranche B is paid-off, the next tranche starts receiving the principal payments and it continues till the senior-most tranche is paid off.

Advantages of MBS

- **Liquidity**: The loans are illiquid assets on the balance sheets of the banks. MBS allows the banks to sell them off to generate cash to be able to make more loans.

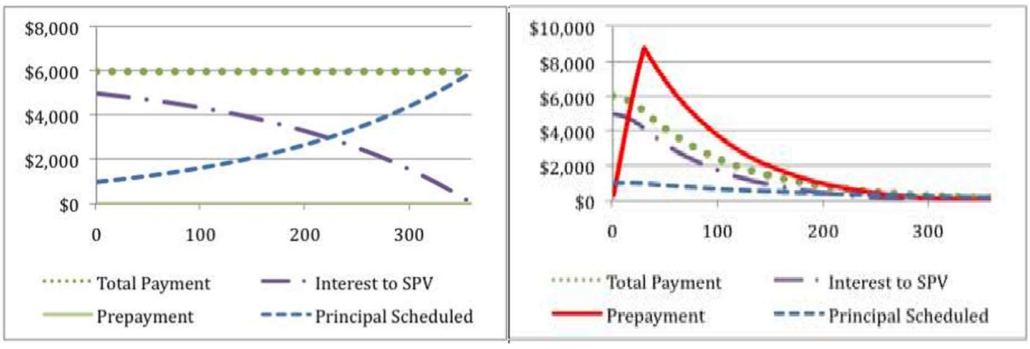
- **Risk Management**: Loans are typically long-term assets and thus have higher durations. The value of the assets becomes highly sensitive to the interest rate movements and increases the banks' risk exposure. MBS allows the banks to make these loans off-balance sheet items and thus reduce their risk exposure.

Prepayment Risk

A borrower can accelerate his payment schedule and pay back the mortgage before the final payment date. This activity is called prepayment and affects the MBS with change in cash flow schedule. Various factors can lead to prepayments- increased income, house turnover and using the sales proceeds to pay back the loan, or refinancing of loans. Refinancing is usually the most prominent reason for prepayments and can be modelled.

The driving factor for refinancing is borrower's incentive, i.e. the difference between the interest rate on the existing mortgage and the prevailing mortgage rates. If rates go down, the borrower can refinance the loan at a lower rate and use the new loan to payback the principal of the old loan. This induces an additional risk factor in MBS compared to a Bond as this would change the expected cash flows for the MBS investor. With principal prepayments, the subsequent interest payments and the life of the cash flows reduces.

1. **MBS Cash Flows with no prepayments** 2. **MBS Cash flows with prepayments**



How does an MBS differ from a bond with a fixed cash-flow schedule?

The option of prepayment lies with the borrower and induces an option cost in the value of the MBS. The prepayments can be voluntary with the borrower refinancing if the prevailing mortgage rates are below the rate on existing mortgage. This is like a call option. The borrower also has a put option with the right to default on their mortgage, leading to the borrower turning over the house to the lender. So, investment in MBS can be viewed as a long position in a non-callable bond with fixed payment schedule and a short position in a call and put option. So, the value of the MBS can be described as

Price of MBS = Price of Non-Callable Bond – Call Option Value – Put Option Value

Prepayments give the principal cash back to the MBS investors earlier than the principal payment schedule. This should increase the present value of the MBS. Then, is prepayment good for the investor?

Prepayments usually occur due to refinancing in low interest rate periods. As the investors get more cash in low interest rate periods, they have to invest more at low interest rates and the gain in value due to lower discount factors is significantly lesser than the loss due to low reinvestments. Thus, the prepayments expose the investors to reinvestment risk.

Regulations

Glass-Steagall Act

The Glass-Steagall Act also known as the Banking Act of 1933, was named after its sponsors, Senator Carter Glass and Congressman Henry Steagall. During the Great Depression, thousands of banks collapsed wiping out depositors' savings. A major cause of the banking sector's problems was the banks' use of deposit funds for funding riskier but more lucrative activities such as securities underwriting and trading.

The Glass Steagall Act aimed to protect depositors from a recurrence of this by splitting banking activities into commercial banking on one hand and investment banking on the other. Commercial banks unlike investment banks and brokerages could raise deposits and benefit from cheap Federal funding but could not pursue investment banking activities. The Act gave banks a year to choose between which of the two activities they would like to pursue.

A famous outcome of the Glass Steagall Act was the breaking up of J.P. Morgan into J.P Morgan (currently J.P. Morgan Chase) and Morgan Stanley.

Volcker Rule

The Volcker rule (named after former Federal Reserve Chairman Paul Volcker) aimed to prohibit certain speculative activities by banks which benefit from public support through the Federal Reserve and the FDIC.

As cited by President Barack Obama while introducing the Volcker rule, "Banks will no longer be allowed to own, invest or sponsor hedge funds, private equity funds or proprietary trading operations for their own profit unrelated to serving their customers."
This implies that banks which benefit from lower cost of funds through access to Federal Reserve funding & federally insured public deposits should not indulge in speculative activities such as running hedge funds, private equity funds or proprietary trading that are unrelated to their core banking business. A major reason for this is to dissuade banks from undertaking excessive risk with public money which can threaten the stability of the financial system necessitating bailouts using taxpayer funds. Another reason is to prevent another unhealthy outcome of bank proprietary investments: misalignment of bank and client interests.

However, the rule allows non-banking financial firms to freely undertake such activities as long as they do not pose systemic risk to the financial system. Such businesses would be free to fail without recourse to taxpayer support.

A much-watered down version of the above rule is expected to become law as part of the Dodd- Frank legislation. The final version restricts banks proprietary investments to 3% of a bank's capital or 3% of an individual fund's capital.

Dodd-Frank Act

The Dodd Frank Act was passed to end the "too big to fail" concept of large banks and create financial stability by increasing accountability and improving transparency in the financial services industry.

The initial bill for the Act was passed on December 2, 2009 by Barney Frank, in the Senate Banking Committee chaired by Chris Dodd. The bill called for the most sweeping overhaul of financial regulation since the Great Depression.

The final Act signed by President Barack Obama on July 21, 2010, contained 90% of the initial proposals and some additional rules such as the Volcker rule which prohibits banks from proprietary trading.

Financial Crises

The Great Depression

The great depression from 1929-33 has been the worst economic crisis that has ever hit the world. The intensity of the crisis was exacerbated by the crisis spreading from the financial economy to the real economy. The crisis started with the stock market crash on October 29, 1929, known as the *Black Tuesday* and soon took shape of economic contraction with industrial output falling 46% in USA in the period of 1929-32, unemployment reaching the heights of 25% and widespread deflation with CPI dropping by 27%.

Asian Currency Crisis

Prior to 1997, the South East Asian countries like South Korea, Thailand, Malaysia and Indonesia had created the "Asian Economic Miracle" with sturdy growth rates of 8-12% for over a decade. These countries were the hotspots for international capital investments due to high growth rates and foreign investments friendly economic policies. This growth was stalled by currency depreciation contagion that rapidly spread through the region and forced the economies and financial markets into a quandary.

The crisis started with Thailand's financial collapse leading to converting Thai Baht from a US$ fixed peg to float. The financial collapse triggered a rapid pullout of foreign investments from other Asian economies as well and nearly all the economies in the region witnessed their currencies plummeting.

Great Financial Crisis

Financial crisis of 2007–08, also called subprime mortgage crisis, severe contraction of liquidity in global financial markets that originated in the United States as a result of the collapse of the U.S. housing market. Although the exact causes of the financial crisis are a matter of dispute among economists, there is general agreement regarding the factors that played a role:

First, the Federal Reserve (Fed), the central bank of the United States, having anticipated a mild recession that began in 2001, reduced the federal funds rate (the interest rate that banks charge each other for overnight loans of federal funds—i.e., balances held at a Federal Reserve bank) 11 times between May 2000 and December 2001, from 6.5 percent to 1.75 percent. That significant decrease enabled banks to extend consumer credit at a lower prime rate (the interest rate that banks charge to their "prime," or low-risk, customers, generally three percentage points above the federal funds rate) and encouraged them to lend even to "subprime," or high-risk, customers, though at higher interest rates (see subprime lending). Consumers took advantage of the cheap credit to purchase durable goods such as appliances, automobiles, and especially houses. The result was the creation in the late 1990s of a "housing bubble" (a rapid increase in home prices to levels well beyond their fundamental, or intrinsic, value, driven by excessive speculation).

Second, owing to changes in banking laws beginning in the 1980s, banks were able to offer to subprime customers mortgage loans that were structured with balloon payments (unusually large payments that are due at or near the end of a loan period) or adjustable interest rates (rates that remain fixed at relatively low levels for an initial period and float, generally with the federal funds rate, thereafter). As long as home prices continued to increase, subprime borrowers could protect themselves against high mortgage payments by refinancing, borrowing against the increased value of their homes, or selling their homes at a profit and paying off their mortgages. In the case of default, banks could repossess the property and sell it for more than the amount of the original loan. Subprime lending thus represented a lucrative investment for many banks. Accordingly, many banks aggressively marketed subprime loans to customers with poor credit or few assets, knowing that those borrowers could not afford to repay the loans and often misleading them about the risks involved. As a result, the share of subprime mortgages among all home loans increased from about 2.5 percent to nearly 15 percent per year from the late 1990s to 2004–07.

Third, contributing to the growth of subprime lending was the widespread practice of securitization, whereby banks bundled together hundreds or even thousands of subprime mortgages and other, less-risky forms of consumer debt and sold them (or pieces of them) in capital markets as securities (bonds) to other banks and investors, including hedge funds and pension funds. Bonds consisting primarily of mortgages became known as

mortgage-backed securities, or MBSs, which entitled their purchasers to a share of the interest and principal payments on the underlying loans. Selling subprime mortgages as MBSs was considered a good way for banks to increase their liquidity and reduce their exposure to risky loans, while purchasing MBSs was viewed as a good way for banks and investors to diversify their portfolios and earn money. As home prices continued their meteoric rise through the early 2000s, MBSs became widely popular, and their prices in capital markets increased accordingly.

Fourth, in 1999 the Depression-era Glass-Steagall Act (1933) was partially repealed, allowing banks, securities firms, and insurance companies to enter each other's markets and to merge, resulting in the formation of banks that were "too big to fail" (i.e., so big that their failure would threaten to undermine the entire financial system). In addition, in 2004 the Securities and Exchange Commission (SEC) weakened the net-capital requirement (the ratio of capital, or assets, to debt, or liabilities, that banks are required to maintain as a safeguard against insolvency), which encouraged banks to invest even more money into MBSs. Although the SEC's decision resulted in enormous profits for banks, it also exposed their portfolios to significant risk, because the asset value of MBSs was implicitly premised on the continuation of the housing bubble.

COVID-19 Crisis

The coronavirus pandemic, which was first detected in China, has infected people in 188 countries. Major events included a described Russia–Saudi Arabia oil price war after failing to reach an OPEC+ agreement that resulted in a collapse of crude oil prices and a stock market crash in March 2020.

Financial Risk - The OECD points out that businesses in many countries have become highly indebted, while the very low cost of borrowing and accommodative monetary policy have contributed to unprecedented corporate debt issuance. Consequently, corporate debt stands at very high levels in many G20 countries. Also, lower-rated credit issued in the form of BBB bonds, non-investment grade bonds, and leveraged loans has risen to elevated levels, the OECD warns, meaning businesses will have little choice but to reduce costs and employment to withstand insolvency pressures.

Stock Market - Overall, stock markets declined over 30% by March; implied volatilities of equities and oil have spiked to crisis levels; and credit spreads on non-investment grade debt have widened sharply as investors reduce risks.

Oil Prices - The COVID-19 pandemic and related confinement measures caused an unprecedented contraction in economic activity and a collapse in demand for oil and oil products. The result is one of the biggest price shocks the energy market experienced since the first oil shock of 1973. Oil prices dipped below USD 20 (Brent crude) a barrel,

losing nearly 70% in value, with storage capacity approaching its limits. The reduction in the demand for travel and the lack of factory activity due to the outbreak significantly impacted demand for oil, causing its price to fall.

Bond Markets - As the economic impact of the coronavirus began to be felt, numerous financial news sources warned of the potential cascade of impacts upon the outstanding $10 trillion in corporate debt. Between mid-February and early March, investors increased the premium, or additional yield, to hold junk bonds by four times the premium demanded of higher credit lenders, indicating increased wariness. During the 2020 stock market crash that began the week of 9 March, bond prices unexpectedly moved in the same direction as stock prices. Bonds are generally considered safer than stocks, so confident investors will sell bonds to buy stocks and cautious investors will sell stocks to buy bonds. As big investors sought to sell, the spread between the prices sellers and buyers wanted has widened. As banks were unable to sell the bonds they were holding, they also stopped buying bonds. As the number of traders fell, the few trades remaining wildly swung the bond prices. Market depth in Treasuries, a measure of liquidity, fell to its lowest level since the 2008 crisis. Also, prices for bond exchange-traded funds began dropping below their net asset values.